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CENTRAL ASIAN AND CAUCASUS REGIONAL FISHERIES AND AQUACULTURE COMMISSION

Report of the

NINTH SESSION OF THE CENTRAL ASIAN AND CAUCASUS REGIONAL FISHERIES AND AQUACULTURE COMMISSION

Almaty, Kazakhstan, 25–26 November 2025

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PREPARATION OF THIS DOCUMENT

This is the final version of the report as approved by the Ninth Session of the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission (CACFish), which was held in Almaty, Kazakhstan, from 25 to 26 November 2025. The report was prepared by the CACFish Secretariat. The material in the appendixes is reproduced as submitted.

ABSTRACT

The CACFish Session took place in Almaty, Kazakhstan, on 25–26 November 2025, hosted by the Fisheries Committee of the Ministry of Agriculture of Kazakhstan. The Session was attended by four Member States: Azerbaijan, Kyrgyzstan, Tajikistan, and Türkiye. Five non-Member States also participated, namely Georgia, Kazakhstan, the Russian Federation, Turkmenistan, and Ukraine. The Commission was briefed on key decisions and recommendations from the respective FAO Regional Conferences, Committees, and Regional Fisheries Bodies, emphasizing the need to ensure that regional priorities are effectively represented in such global and regional decision-making platforms and processes. The Secretariat reported on administrative and financial matters, noting improved compliance by one Member State but persistent arrears from two others, which remain a major challenge to financial sustainability. Membership expansion was highlighted as promising, with additional countries expected to join as accession processes advance. Technical discussions focused on regionalizing FAO Guidelines for Sustainable Aquaculture, zoning and site selection, developing a fisheries data collection system, and adopting Best Practice Guidance for Occupational Safety and Health, approved as a non-binding instrument. The Commission acknowledged its growing range of activities, expanded technical scope, and significant achievements during the previous intersessional period, underscoring its role in addressing regional challenges. The Commission endorsed an autonomous annual budget of USD 180,000 for 2025–2026, reaffirmed financial support for Member and invited non-Member participation, approved revisions to the Rules of Procedure, and adopted its work programme for 2025–2027.

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Opening of the Session

1. The Ninth Session of the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission (CACFish) was held in Almaty, Kazakhstan, from 25 to 26 November 2025. The event was hosted by the Fisheries Committee under the Ministry of Agriculture of the Republic of Kazakhstan.
2. As Chair, Zhalilbek Toimatov opened the Session and welcomed participants.
3. The Session was attended by four CACFish Member States: Azerbaijan, Kyrgyzstan, Tajikistan, and Türkiye. In addition, five non-CACFish Member States participated: Georgia, Kazakhstan, the Russian Federation, Turkmenistan and Ukraine. The list of participants is provided in Appendix B.
4. A welcoming statement on behalf of the host country was delivered by Mr Serik Sermagambetov, Chair of the Fisheries Committee. He highlighted Kazakhstan's abundant water resources and its commitment to aquaculture as a rapidly growing global sector. He addressed the adoption of the Fisheries Development Program until 2030 and the recent Law on Aquaculture, emphasizing that legislative improvements and state support create opportunities for innovation and growth.
5. Haydar Fersey, Senior Fishery and Aquaculture Officer of FAO and Secretary of the Commission, welcomed participants and underscored CACFish's role as the leading regional platform for addressing inland fisheries and aquaculture challenges. He highlighted its contribution to building institutional capacity, promoting sound management practices, and fostering the exchange of experiences and lessons learned among Member and observer States.

Adoption of the agenda

6. The Commission adopted the agenda without changes (Appendix A). The list of documents which were placed before the Session is presented in Appendix C.

Election of Chair and Vice-chairperson

7. Makhmadullo Odilov (Tajikistan) and Rafiq Mirzayev (Azerbaijan) were elected as Chairperson and First Vice-Chairperson of the Commission, respectively. The Commission did not elect a Second Vice-Chairperson, as had occurred on several previous occasions.
8. The incoming Chairperson expressed appreciation to the outgoing Chairperson and First Vice-Chairperson for their leadership and valuable contributions to the work of the Commission.

Main decisions and recommendations of the respective FAO Regional Conferences, Committees and regional fisheries bodies

9. The Secretariat introduced the agenda item based on the working document CACFish/IX/2025/2, The Commission took note of the key decisions and recommendations of the following meetings:
 - Thirty-sixth Session of the FAO Committee on Fisheries (COFI), Rome, Italy, 8–12 July 2024;
 - Thirty-fourth Session of the FAO Regional Conference for Europe, Rome, Italy, 14–17 May 2024;
 - Thirty-second Session of the European Inland Fisheries and Aquaculture Advisory Commission (EIFAAC), Pula, Croatia, 9–11 October 2024; and
 - Eighth Session of the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission, Baku, Azerbaijan, 19–20 December 2023.
10. Two pre-recorded video presentations were shown: Alice McDonald, Secretary of COFI, reported on the Thirty-sixth Session, and Mr Raymon Van Anrooy, Secretary of EIFAAC, reported on the Thirty-second Session. The remaining presentations were delivered by the Secretariat.
11. The Commission emphasized the importance of COFI and its sub-committees in supporting its mandate and encouraged active participation by States within its area of competence to ensure regional priorities and perspectives are effectively reflected.

Secretariat report on administrative, operational, and financial issues

12. Haydar Fersoy, CACFish Secretary briefed the Commission on administrative, operational, and financial matters, as outlined in document CACFish/IX/2025/3. The presentation included an overview of activities carried out during the intersessional period, highlighting their key outcomes, messages, and recommendations from technical regional events such as workshops and expert group meetings. The Commission expressed its appreciation for the substantial work undertaken during this period.

13. The Secretariat provided details on the overall budget status and financial expenditures incurred during the intersessional period. The payment situation showed improvement compared to previous years, primarily due to Kyrgyzstan settling nearly all outstanding arrears and consistently paying its annual contributions. However, two Member States have not made any contributions since joining, leaving the overall arrears level largely unchanged.

14. Despite these persistent arrears, the funds accumulated over recent years will enable the Commission to maintain the same level of work programme activities as in the previous intersessional period, which recorded a historic number of initiatives. This demonstrates the Commission's ability to sustain a high level of programme delivery and performance, even under conditions of partial compliance with contribution obligations.

15. The Commission noted that its work programme has primarily consisted of technical activities since its establishment. These include regional workshops and specialized expert group meetings. Major expenditure items covered travel costs for delegates and experts, as well as expenses related to organizing training events. Consultancy costs remained relatively limited, as the Secretariat undertakes some technical work directly.

16. Türkiye expressed concern that these long-standing arrears undermine the Commission's effectiveness and financial stability, limiting its ability to expand activities. Persistent arrears were identified as a major obstacle to achieving financial sustainability. Members called for practical steps to resolve the issue, emphasizing that timely contributions from all Members are essential for shared responsibility and long-term viability. The CACFish Secretariat was tasked with following up on this matter with the respective Member States and exploring possible solutions, including the development of an arrears payment plan.

17. Building on this financial stability, the Commission acknowledged that accumulated reserves will allow the continuation of a robust work programme at the same level as the previous intersessional period, which achieved a record number of initiatives. This underscores the Commission's resilience and capacity to deliver strong technical outputs despite financial constraints.

18. The Commission noted with appreciation the significant efforts undertaken during the previous intersessional period by the CACFish Chair, Vice-Chair, and Secretariat to advance membership expansion, as requested at the last session. These efforts included sending official membership invitation letters, informing prospective countries about accession procedures and contribution schemes, and engaging in active outreach and lobbying with relevant authorities. FAO supported these initiatives by holding bilateral meetings with high-level representatives of countries considering membership, reiterating its call for them to join the Commission. The Commission further noted with appreciation that additional Member States are expected to join in the coming years.

19. The Commission was informed that membership processes are progressing in several countries. One country has completed internal consultations and formalities and is now awaiting final accession approval. Two other countries are advancing with internal procedures, showing encouraging progress toward joining. The Commission recognized these developments as a direct outcome of sustained outreach efforts and ongoing partnerships with countries engaged since the Commission's pre-establishment phase.

20. The Commission reiterated that expanding its membership base would strengthen its mandate and enhance regional cooperation in sustainable fisheries and aquaculture management and development. It encouraged continued engagement to facilitate these processes and maintain momentum toward enlargement. The Secretariat was tasked with closely monitoring and supporting ongoing accession procedures.

Report on recommendations and activities of the Technical Advisory Committee

21. The Secretariat briefed the Commission on the activities of the Technical Advisory Committee (TAC) during the 2023–2025 intersessional period. Rafiq Mirzayev, speaking on behalf of the TAC Chair Country, Azerbaijan, reported on the main decisions and recommendations from the Seventh Meeting of TAC, which was held in Baku, Azerbaijan, on 13–14 May 2025. It was noted that the meeting discussed and provided technical advice on the following items:

- regionalization of FAO Guidelines for Sustainable Aquaculture (GSA);
- zoning and site selection in aquaculture;
- development of a CACFish data collection system for fishing gear use and catches; and
- Best Practice Guidance for Occupational Safety and Health (OSH) in Inland Aquaculture and Fisheries.

22. The Commission acknowledged that the FAO Guidelines for Sustainable Aquaculture (GSA) provide a comprehensive framework for inclusive planning and decision-making, which is essential for addressing challenges posed by rapid aquaculture growth in Central Asia and the Caucasus. The adoption of GSA principles in the CACFish region would help prevent unsustainable practices and associated environmental impacts, while promoting responsible intensification. To advance implementation, the Commission emphasized the need to:

- expand training and capacity-building programmes;
- strengthen communication and awareness across the sector;
- encourage formal stakeholder endorsements; and
- develop national and regional strategies and action plans aligned with GSA principles.

23. The Commission noted that poorly planned aquaculture development continues to create environmental and social risks, thereby reducing economic benefits. This underscores the urgent need for spatial planning tools to establish aquaculture zones based on environmental capacity, social considerations, and economic potential. Within these zones, applying technical site selection criteria is essential to ensure farms operate in suitable and sustainable locations. The Commission emphasized the importance of a focused regional approach, including frameworks and guidelines to inform national management and production practices, protocols and standards.

24. The Commission examined the development of the CACFish data collection framework for recording fishing gear usage and catch statistics. The discussion referenced the Regional Expert Workshop held in Istanbul in January 2025 and the preparatory questionnaire distributed to member countries. The framework was deemed technically robust and responsive to regional priorities; however, its current application is confined to inland fisheries.

25. The session emphasized the importance of strengthening national fisheries data systems to ensure compatibility with the CACFish framework. Several countries will need to establish foundational fisheries information systems, which will require considerable investment. The Commission agreed to further refine the technical framework and initiate pilot projects in selected fisheries, prioritizing commercial operations over artisanal sectors.

26. Regarding the Best Practice Guidance for Occupational Safety and Health (OSH) in Inland Aquaculture and Fisheries—originally developed as the CACFish Safety Code of Practice and reviewed at the Eighth Session—the Commission noted that the Regional Workshop held in Bishkek, Kyrgyzstan, on 25–26 February 2025, reviewed the Code and renamed it as Best Practice Guidance with amendments, including enhanced provisions for aquaculture. The Guidance aims to reduce risks in these high-hazard

sectors and promote proactive worker health and safety through practical, risk-based measures aligned with international standards. The Commission acknowledged the action steps proposed by the workshop: raising awareness, developing OSH standards and training packages, establishing injury and disease reporting systems, and supporting research on social protection and insurance. The Commission approved the Best Practice Guidance as a non-binding instrument, provided in Appendix D.

27. The Commission expressed appreciation for the high-quality technical work undertaken by TAC and the valuable advice provided. It noted the importance of continued collaboration to ensure effective implementation of these recommendations.

Commission budget and Member contributions for 2025 and 2026

28. The Commission examined the proposed autonomous budget for the biennium 2025–2026, as set out in working document CACFish/IX/2025/5, along with the related estimated expenditures and agreed Member contributions. Following its review, the Commission approved the two-year budget, presented in Appendix E.

29. The Commission noted that new Members may join during the 2025–2026 biennium. A Member joining in 2025 will contribute to the budget for both 2025 and 2026, whereas a member joining in 2026 will contribute only for that year. The total annual budget (USD 180 000) will remain unchanged; however, individual contributions will be recalculated to include new Members for the applicable year(s). This adjustment will proportionally reduce the payment amounts of existing Members for the year in which a new Member joins.

30. The Commission noted that only one regional workshop was held exclusively in English during the 2023–2025 work programme. It agreed to continue organizing a limited number of technical activities—such as workshops and expert group meetings—only in English, in line with the decision taken at its Eighth Session (2023). This approach reflects the high cost of interpretation and the fact that English is the Commission’s sole official language.

31. The Commission decided to continue providing financial support to facilitate the participation of Member States and invited non-Member States in its events and those of its subsidiary bodies. This strategy promotes broad regional representation, fosters partnerships, and strengthens cooperation. The active involvement of non-Member States has consistently enhanced regional collaboration and increased the Commission’s visibility, benefiting all participants. Under the approved 2025–2027 work programme, as outlined in paragraph 35, financial support will cover up to three participants from each Member State and one from each invited non-Member State for regional meetings and activities. Such engagement was considered essential to the Commission’s work plan.

32. Türkiye reaffirmed CACFish’s mandate and leadership in regional aquaculture, emphasizing the need to strengthen institutional and financial sustainability, align activities with blue transformation principles, and maintain a realistic, results-oriented budget. It called for timely payments, efficient resource use, and constructive dialogue on financial challenges, while urging alignment with FAO priorities such as climate adaptation, blue economy, data-driven management, and sustainable aquaculture. Türkiye reiterated its commitment to species diversification, integrated aquaculture systems, and gender equality, and proposed closer collaboration with EIFAAC. It also requested the Secretariat to explore options for securing external funding for specific projects through voluntary contributions or global mechanisms such as the Global Environment Facility (GEF) and climate change-related funding sources.

33. The Commission stressed the importance of timely nomination of participants with appropriate technical expertise and language skills to ensure effective engagement in meetings and activities, particularly expert group work. Prompt responses to surveys and requests for data and information are also essential to support the Commission’s work programme and associated outcomes.

Review of the Rules of Procedure of the Commission

34. The Secretariat introduced the revised Rules of Procedure, shown with tracked changes in document CACFish/IX/2025/6, and explained that they underwent a moderate review in consultation with FAO divisions, as requested by the Commission at its Eighth Session. The revision was based on the 2022 amended Rules of Procedure of the General Fisheries Commission for the Mediterranean, another FAO Article XIV body and the text received FAO legal clearance. Key changes included correcting a typographical error in Rule XVI, citing reference sources and integrating relevant provisions, as well as revising language to align with FAO standards and improve clarity and consistency. After review, the Commission approved the revised Rules of Procedure without further amendments. The amended Rules of Procedure adopted by the Commission are provided in Appendix F.

Work programme for 2025–2027 inter-sessional period

35. The Commission reviewed the work programme activities proposed by the Technical Advisory Committee at its seventh meeting (2025) and agreed on the following work programme for the 2025–2027 intersessional period, incorporating additional activities identified by the Commission. It also endorsed the national-level work programme activities. Furthermore, it was agreed that if a new Member joins during this period, that country will receive financial support of USD 25,000 from CACFish to implement a national-level activity.

#	Activity	Timeline	Estimated budget (USD)	Place
i.	Regional expert meeting for the development of a CACFish data collection system for fishing gear use and catches	May 2026	30 000	Azerbaijan
ii.	Regional workshop on seed supply for small-scale aquaculture	Feb 2026	30 000	Türkiye
iii.	Expert meeting on fish stocking into inland water bodies	2026	30 000	Georgia
iv.	Workshop on sturgeon aquaculture and conservation	2026	30 000	Tajikistan
v.	Public awareness on fish consumption and health benefits of fish consumption (video, etc.)	2026	20 000	N/A
vi.	Regional workshop on enhancement of the aquaculture value chain	2027	30 000	Azerbaijan
vii.	Fish health management - preparedness and response to diseases of fish	2027	30 000	Uzbekistan
viii.	Workshop on site selection and zoning in aquaculture	2027	30 000	TBC
ix.	A special aquatic biotechnology session at the 3rd International Congress on Biotech Solutions for Sustainability (Gebze, Türkiye, 2026)	2026	20 000	Türkiye
x.	A regional workshop on pike perch aquaculture	Sept 2026	25 000	Kazakhstan
xi.	Attendance of one representative at the 37th Session of the Committee on Fisheries – restricted to members only	2026	12 000	Rome, Italy
xii.	Eight Meeting of the TAC	May 2027	30 000	Kyrgyzstan
xiii.	Attendance of one representative at EIFAAC International Symposium	2026	18 000	Riga, Latvia
xiv.	A national-level study for Azerbaijan	2026–2027	20 000	
xv.	A national-level study for Kyrgyzstan	2026–2027	30 000	
xvi.	A national-level study for Tajikistan	2026–2027	30 000	
xvii.	A national-level study for Türkiye	2026–2027	20 000	

Any other matters

36. The Commission expressed its sincere appreciation to the host country for the warm hospitality and excellent arrangements provided throughout the Session.

Date and place of the next session

37. Tajikistan kindly offered to host, subject to a final confirmation by the Government, the Tenth Session of the Commission in October 2027.

Adoption of the report

38. The report was adopted on 26 November 2025.

APPENDIX A: Agenda

25 November 2025	
<i>Morning Session</i> <i>09.30–13.00</i>	1. Opening of the session 2. Adoption of the agenda 3. Election of Chair and Vice-chairperson 4. Main decisions and recommendations of the respective FAO Regional Conferences, Committees and Regional Fisheries Bodies (document: CACFish/IX/2025/2) 5. Secretariat report on administrative, operational, and financial issues (document: CACFish/IX/2025/3)
Lunch break	
<i>Afternoon Session</i> <i>14.00–17.00</i>	6. Report on recommendations and activities of the Technical Advisory Committee (document: CACFish/IX/2025/4) 7. Commission budget and Member contributions for 2025 and 2026 (document: CACFish/IX/2025/5) 8. Review of the Rules of Procedure of the Commission (document: CACFish/IX/2025/6) 9. Work programme for 2025-2027 inter-sessional period 10. Any other matters 11. Date and place of the next session
26 November 2025	
<i>Morning Session</i> <i>9.00–12.00</i>	Free (Secretariat to prepare the session report)
<i>Afternoon Session</i> <i>15.30–17.00</i>	Adoption of the session report

APPENDIX B: List of participants

MEMBER COUNTRIES

AZERBAIJAN

Rafiq Mirzayev

Head of the Aquaculture Department, Ministry of Agriculture, Republic of Azerbaijan

Leyla Taghiyeva

Senior Specialist of the Fisheries Department, Ministry of Agriculture, Republic of Azerbaijan

KYRGYZ REPUBLIC

Zhalilbek Toimatov

Director of the Department of Fisheries Industry Complex under the Ministry of Water, Agriculture and Processing, Kyrgyz Republic

Kuanych Satybekov

Chef Specialist, Department of Fisheries Industry Complex, Kyrgyz Republic

TAJIKISTAN

Mazmadullo Odilov

Head of the Livestock, Poultry, Fisheries, and Beekeeping Sector of the Ministry of Agriculture, Republic of Tajikistan

Parviz Chabborzoda

Director of the Mokhii Tojikiston branch in Sughd Region, Republic of Tajikistan

TÜRKİYE

Nimet Kavuz

Coordinator responsible for CACFish, Aquaculture Department of the General Directorate of Aquaculture and Fisheries, Ministry of Agriculture and Forestry, Türkiye

Burçak Yüksel

Expert, Aquaculture Department of the General Directorate of Aquaculture and Fisheries, Ministry of Agriculture and Forestry, Türkiye

INVITED COUNTRIES

GEORGIA

Shorena Mchedlidze

First Category Senior Specialist of Fisheries Development Division, Agriculture and Rural Development Policy Department, Georgia

KAZAKHSTAN

Serik Sermagambetov

Chairperson of the Fisheries Committee of the Ministry of Agriculture, Republic of Kazakhstan

Yerdaulet Kakimzhanov

Chief Expert of the Fisheries Committee of the Ministry of Agriculture, Republic of Kazakhstan

Serik Beisimbayev

Head of the Nura-Sarysu Interregional Basin Inspectorate of Fisheries of the Fisheries Committee of the Ministry of Agriculture, Republic of Kazakhstan

Ainur Kaldayeva

Deputy Head of the Balkhash-Alakol Interregional Basin Fisheries Inspection, Republic of Kazakhstan

Aidana Tlepbay

Chief Specialist of the Department of Fisheries, Monitoring, and Digitalization, Republic of Kazakhstan

Zhansulu Urganishbayeva

Ichthyopathologist, Head of Laboratory at “Ospanova Development” LLP, Republic of Kazakhstan

Erdaulet Kakimjanov

Chief Expert of the Fisheries Committee of the Ministry of Agriculture, Republic of Kazakhstan

Arailym Umirtayeva

Researcher, “Scientific and Production Center for Fisheries”, Republic of Kazakhstan

Murat Duisenbayev
Fish farmer, Republic of Kazakhstan

RUSSIAN FEDERATION

Anastasia Podolyan
Representative of the
Federal Agency for Fisheries and Permanent
Mission of the
Russian Federation to FAO
and other UN Agencies in Rome

Margarita Kogai
Lead Specialist of the Department, Russian
Federal Research Institute of Fisheries and
Oceanography (VNIRO), Russian Federation

UKRAINE

Tetiana Yakovlieva
Head of the Aquaculture Division, State
Agency for the Development of Melioration,
Fisheries and Food Programs, Ukraine

TURKMENISTAN

Nurgozel Aymammedova
Chief Specialist of the Department for
Development of Fisheries Industries, Republic
of Turkmenistan

FAO/CACFish SECRETARIAT

Haydar Fersoy
Secretary of CACFish, Senior Fishery and
Aquaculture Officer Regional Office for
Europe and Central Asia, FAOREU

Onur Hasaltuntas
Government Provided Staff - Fisheries and
Aquaculture Expert, FAOSEC

Mairam Sarieva
CACFish Secretariat Support Coordinator,
FAOSEC

Aizhan Karabayeva
Project Coordinator, FAOKZ

Aisha Sakhmetova
Operations Support Specialist, FAOKZ

APPENDIX C: List of documents

Working documents

CACFish/IX/2025/1	Provisional Agenda
CACFish/IX/2025/2	Main decisions and recommendations of the respective FAO Regional Conferences, Committees and Regional Fisheries Bodies
CACFish/IX/2025/3	Secretariat report on administrative, operational, and financial issues
CACFish/IX/2025/4	Report on recommendations and activities of the Technical Advisory Committee
CACFish/IX/2025/5	Commission budget and Member contributions for 2025 and 2026
CACFish/IX/2025/6	Review of the Rules of Procedure of the Commission

Information documents

CACFish/IX/2025/Inf.1	Provisional list of documents
CACFish/IX/2025/Inf.2	Provisional list of participants
CACFish/IX/2025/Inf.3	Report of the Eighth Session of the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission, Baku, Azerbaijan, 19–20 December 2023
CACFish/IX/2025/Inf.4	Report of the Thirty-sixth Session of the Committee on Fisheries
CACFish/IX/2025/Inf.5	Report of the 34th Session of the Regional Conference for Europe (Rome, Italy, 14–17 May 2024)
CACFish/IX/2025/Inf.6	Report of the Thirty-second Session of the European Inland Fisheries and Aquaculture Advisory Commission – Pula, Croatia, 9–11 October 2024

Reference documents

CACFish/IX/2025/Ref.1	Agreement on the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission
CACFish/IX/2025/Ref.2	Financial regulations of the Commission

APPENDIX D: Best practice guidance for Occupational Safety and Health (OSH) in inland aquaculture and fisheries¹

1. OBJECTIVES AND SCOPE

Objectives

The overall objective of this *Best Practice Guidance for Occupational Safety and Health (OSH) in Inland Aquaculture and Fisheries* (OSH Best Practice / or “Guidance” in short) is to promote a preventive OSH culture in the inland Aquaculture / Fisheries Sectors. In particular, it aims to protect and promote worker OSH and prevent occupational injuries and diseases by:

- (a) raising awareness of the risks associated with inland aquaculture/ fisheries and how these risks can be effectively managed and controlled;
- (b) creating a healthy and safe working environment in inland aquaculture/ fisheries operations; and
- (c) promoting an OSH culture characterized by positive enabling attitudes and behaviour towards OSH issues in inland aquaculture/ fisheries operations.

Scope

The State of World Fisheries and Aquaculture (FAO, 2024) estimated that of the total production of aquatic animals in 2022, 62 percent was harvested in marine areas (69 percent from capture fisheries, 31 percent from aquaculture) and 38 percent in inland waters (84 percent from aquaculture, 16 percent from capture fisheries). The aquaculture sector has grown dramatically over the past five decades. In 2024 aquaculture has surpassed for the first time in history fisheries production accounting for more than half (51 percent) of the world’s fish food supply. FAO estimated that 61.8 million people were engaged in the primary sector of fisheries and aquaculture in 2022. Many more are also employed along the supply chain. Some 36 percent were employed in aquaculture and 29 percent in inland fisheries globally. Overall, women accounted for 24 percent of those engaged in the primary sector (25 percent in aquaculture and 28 percent in inland fisheries).

It is becoming increasingly apparent that fisheries and aquaculture are high-risk sectors associated with an increasing incidence of occupational injuries, diseases and fatalities (Ngajilo and Jeebhay, 2019). Preventive occupational safety and health measures are lagging behind other sectors. As aquatic production is projected to increase exponentially, the likelihood of occupational safety and health challenges is likely to increase and threaten the sustainability of the industry, unless these are addressed in a wholistic manner (Cavalli, 2019; Watterson, 2020).

Aquaculture is defined as ‘*the cultivation of aquatic organisms in controlled aquatic environments involving interventions in the rearing process to enhance production.*’. Aquaculture may be either marine-based (also called mariculture) or land-based. This OSH Guidance will focus only on land-based fish farms (indoor and/or outdoor) located on or near small water-bodies (e.g. ponds, small lakes) as well as large water bodies (e.g. streams, lakes, reservoirs, dams).

The International Labour Organization (ILO) “Future of work in Aquaculture in the context of the rural economy” report states that:

aquaculture is an important source of income and livelihoods for many rural communities, both coastal and inland. The sector has grown dramatically over the past five decades and now accounts for half of the world’s fish food supply. It provides direct employment to more than 20 million people, with many more people employed along the supply chain. With the growing world population and environmental pressures, aquaculture is increasingly recognized as holding potential for sustainably addressing the challenges of food and nutrition security (ILO, 2019).

¹ The word inland applies to both fisheries and aquaculture.

Another important source of food and employment is inland fisheries. Inland fisheries include any "any activity conducted to extract fish and other aquatic organisms from inland waters" that may either be small water-bodies (e.g. ponds, small lakes, low-order streams) as well as large water bodies (e.g. streams, lakes, reservoirs, dams, canals). This activity generally requires the use of fishing boats/vessels and fishing nets (e.g. handline, purse seine, longline, traps, cages).

Inland aquaculture and fisheries activities are often linked and workers face similar hazards when engaged in work related to these sectors. Therefore, these two complementary sectors and the OSH issues are addressed together in this Guidance. While acknowledging the importance of aquaculture/inland fisheries for food production, there is also the recognition that it can also be a hazardous sector for these working populations. This OSH Guidance is intended to be widely applicable and relevant to all inland aquacultural and fisheries enterprises, irrespective of size, and designed for the protection of all employers as well as workers, irrespective of their employment status.

This OSH Best Practice Guidance provides best practice guidance for a number of workplace issues, based on common operational processes and activities in inland aquaculture/fisheries, and provides a starting point for identifying and controlling hazards in the workplace.

This OSH Best Practice Guidance should be interpreted and applied in accordance with national legislative requirements and their institutional requirements. Employers must take every reasonable precautionary measure to protect the health and safety of all people at the workplace.

It is envisaged that each individual organization uses the information in this OSH Guidance as the basis for designing a comprehensive workplace OSH management program. This includes but is not limited to a health and safety policy, evaluating and controlling hazards, providing OSH information, education and training, monitoring environmental exposures and the health of workers, and collecting data on occupational injuries and diseases on an ongoing basis in order to monitor the effectiveness of workplace control measures.

Workers in inland aquaculture/ fisheries should be provided with the information necessary to ensure their health and safety. This Best Practice Guidance is therefore presented as part of an OSH awareness plan, rather than a comprehensive definitive document of best work practices in these sectors. It is a requirement that all stakeholders with specific responsibilities for OSH should have immediate access to this Guidance, and that it should be made readily available to all workers in so far as is reasonably practicable.

2. WORK PROCESSES, OCCUPATIONAL HAZARDS AND HEALTH EFFECTS IN INLAND AQUACULTURE/FISHERY WORKERS

Work processes and occupational hazards

Inland aquaculture and fisheries are characterised by certain work processes that share common principles. It is the nature of the work, the work process or the task that generates the particular hazard as is evident in **Table 1** below. Both sectors share common hazards such as safety hazards causing physical injuries, drowning, exposure to chemical contaminants and musculoskeletal injuries. The hazards in inland aquaculture are often associated with more controlled environments while inland fisheries also have to deal with natural and unpredictable weather conditions.

Table 1. Main work processes and examples of associated occupational hazards in inland aquaculture and fisheries

Work process	Occupational hazards in aquaculture	Occupational hazards in fisheries
Hatchery operations	– Falling live tank lids causing physical injuries. – Exposure to noise from aeration pumps and other machinery. – Exposure to disinfectants and antibiotics causing skin or respiratory issues. – Exposure to reproductive hormones. – Repetitive strain injuries from sorting and handling fry.	– Physical injuries from handling wild juvenile fish. – Risk of skin infections from diseased fish or contaminated water. – Psychosocial stress from working alone for long periods of time.
Feeding and rearing	– Slippery and wet surfaces increasing the risk of slips and falls due to absent railings on raceway wall edges. – Exposure to fish-feed causing respiratory allergy and asthma. – Exposure to contaminated water leading to skin infections. – Manual lifting of heavy feed bags causing musculoskeletal injuries.	– Strains or sprains from handling feed or traps. – Slippery decks increasing risk of falls during feeding operations. – Electric shocks during electrofishing. – Sunburn, dehydration, or heatstroke when working outdoors.
Fishing vessel operations, catching and harvesting	– Cuts from sharp tools and fish fins during handling. – Risk of drowning in pond or cage environments. – Decompression illness due to diving. – Hearing loss due to noisy machinery. – Repetitive movements causing musculoskeletal disorders. – Heavy lifting leading to musculoskeletal strain.	– Physical injuries from working in rugged terrains or submerged environments. – Injuries from fishing hooks, nets, or sharp fish fins. – Risk of physical harm from animal attacks (e.g., snakes, insects near water). – Crush injuries from malfunctioning winches or mechanical parts on boats. – Fire or explosion risks from improperly maintained fuel systems. – Drowning due to capsizing vessel or falling into water. – Hearing damage from prolonged exposure to engine noise. – Musculoskeletal strain from repetitive casting and hauling heavy nets.

Work process	Occupational hazards in aquaculture	Occupational hazards in fisheries
Water quality management	– Electrical hazards from aeration pumps or filtration systems. – Asphyxiation from poor ventilation in confined spaces and enclosed facilities. – Burns or chemical splashes from pH adjusters or disinfectants. – Exposure to hazardous chemicals such as ammonia, chlorine and formaldehyde.	– Exposure to runoff or agricultural chemicals during monitoring activities. – Skin and respiratory irritation and infections from contact with polluted water.
Disease management	– Needle-stick injuries from vaccinating fish. – Allergic reactions to chemicals or antibiotics. – Skin and other infections from handling fish.	– Contact with parasites or pathogens when managing wild fish stocks. – Physical harm from invasive species or harmful aquatic organisms.
Processing and packaging	– Risk of cuts or amputations from machinery such as fish gutting devices. – Cold environments in processing plants leading to hypothermia or joint problems. – Allergy, asthma and dermatitis from processing fish. – Repetitive motion injuries from sorting and filleting fish.	– Cuts or amputations from handling knives or fish cleaning equipment. – Poor handling practices causing musculoskeletal injuries. – Exposure to cold environments, leading to hypothermia or joint problems. – Allergy, asthma and dermatitis from processing fish. – Repetitive motion from manually sorting fish causing musculoskeletal strain.
Transport	– Slips and falls on wet surfaces during loading or unloading. – Vehicle-related injuries during transportation of aquaculture products. – Heavy lifting of fish crates causing back strain.	– Slips and falls on wet surfaces during unloading at landing sites. – Risk of accidents during transport in poorly maintained vehicles. – Heavy lifting of fish crates causing back strain.
Infrastructure and gear maintenance	– Cuts or abrasions while repairing nets, cages, or tanks. – Falls from heights when repairing tanks, cages, or other structures – Injuries from forklift and crane overturns. – Electrical hazards when handling faulty equipment or submerged systems. – Fires or explosions from oxygen cylinders. – Chemical exposure during cleaning and disinfecting tanks, boats or equipment (e.g., paint or anti-fouling chemicals). – Strains from heavy lifting or awkward postures when handling gear.	– Injuries from repairing fishing gear or nets. – Falls from unstable platforms or while handling large nets or traps.

Occupational hazards and health effects

Depending on the frequency and duration of exposure as well as the nature of a particular hazard, this can have different effects. If the incident is severe this can result in sudden death, serious injury/disease, or a minor injury/illness. **Table 2** highlights the diverse nature of occupational hazards and their wide-ranging adverse health impacts resulting in chronic disease affecting different organ systems, such as occupational allergies, lung or kidney disease, nerve or musculoskeletal disorders, cancers, reproductive problems and birth defects.

Table 2. Classification of common hazards in inland aquaculture/fisheries and their adverse health effects

Hazard category	Causative agents or work processes	Adverse health effects
Safety	Slips and trips, falls (height, overboard)	Sprains, strains, fractures
	Needle-sticks	Injuries
	Transport and trucking	Injuries
	Unprotected machinery/chainsaws	Injuries, cuts, amputations
	Electricity	Electrocution
	Entanglement, underwater entrapment	Drowning
	Diving	Decompression illness, drowning, oxygen poisoning
	Battery explosions, fire	Burns
	Firearms (predator control)	Injuries
	Wet surfaces	Trips, slips and falls
	Tractor power take-offs (PTO)	Injuries
	Pressure (or power) washers	Injuries, electrocution, musculoskeletal injuries
	Boating	Drowning
Physical	Confined spaces/poor ventilation	Oxygen deprivation (unconsciousness)
	Extreme temperatures	Hypothermia, heat stress, dehydration, kidney disease
	Vibration	Backache, vibration white finger
	Solar radiation	Sunburn, keratitis, cataract, pterygium, blindness
	Noise	Hearing loss
	Water immersion	Drowning, hypothermia, skin maceration
Chemical	Formaldehyde, disinfectants	– Eye, airways and skin irritation or allergic effects causing rhinitis, asthma, dermatitis, – Cancer (formaldehyde)
	Antibiotics	Altered airway and gut flora, antibiotic resistance, allergies
	Reproductive hormones	Hormonal disturbances
	Water quality tests, pH adjustment chemicals	Eye, airways and skin irritation causing rhinitis, asthma, dermatitis
	Toxic gases (fish decomposition, chemicals)	Asphyxiation, lung tissue damage
	Anaesthetics	Severe eye and skin irritation
	Preservatives	Eye, airways and skin irritation or allergic effects causing rhinitis, asthma, dermatitis

Hazard category	Causative agents or work processes	Adverse health effects
	Silicone-based defoaming agents	Severe eye and skin irritation
	Styrene (polystyrene, fibreglass products)	– Eye, airways and skin irritation or allergic effects causing rhinitis, asthma, dermatitis – Nerve damage
	Agrichemicals (pesticides), antifoulants	– Eye, airways and skin irritation or allergic effects causing rhinitis, asthma, dermatitis – Nerve, kidney, blood organ damage, reproductive and endocrine effects, birth defects, cancers
	Welding fumes	– Irritation of the eyes, nose, upper and lower airways and skin causing rhinitis, asthma – Cataracts
Biological	Sharp teeth, spines	Bites, cuts, punctures, infections
	Poisonous animals/insects	Bites, toxic reactions, allergies
	Fish proteins, fish feed dust	Allergy, rhinitis, asthma, dermatitis
	Microorganisms (e.g. bacteria, viruses, mould, parasites)	– Infections – Allergies, rhinitis, asthma
	Endotoxin	Organic dust toxic syndrome, airway irritation, rhinitis, asthma
Ergonomic	Heavy lifting, awkward work postures, prolonged standing, repetitive movements	Low back, neck, shoulder and wrist pain
Psychosocial	Large fish kills, remote work, abusive environment, sleep deprivation, long shifts	Work-related stress, anxiety, depression

Source: Adapted from Ngajilo, D. & Jeebhay, M.F. 2019. Occupational injuries and diseases in aquaculture. *Aquaculture*. 507:40–55. doi:10.1016/j.aquaculture.2019.03.053

3. IDENTIFYING AND MANAGING OCCUPATIONAL HAZARDS IN INLAND AQUACULTURE/ FISHERIES - A RISK-BASED APPROACH

Introduction

It is the responsibility of governments to establish the appropriate legal framework and policies to enforce them to ensure a healthy and safe working environment in inland aquaculture and fisheries. This is done through collaboration between relevant government ministries and stakeholders, which include employer, worker and community-based organisations (where appropriate). It is the primary responsibility of the employer to address occupational safety and health (OSH) risks associated with processes and tasks all along the value chain.

Occupational safety and health risks emanating from inland aquaculture and fisheries are best managed using the OSH Risk Management Cycle. It involves a systematic process to identify, assess and control workplace hazards and to monitor their effectiveness on an ongoing basis. At the heart of this process is commitment of management to drive this process, in consultation with workers and their representatives, supported by competent OSH professionals that provide appropriate technical expertise. This cyclical process ensures a proactive and dynamic approach to workplace health and safety, thereby promoting continuous improvement and compliance with laws, regulations, policies and codes of practice (see next section) for this sector. The OSH Risk Management Cycle (see **Figure 1**) consists of four steps:

- 1. Hazard identification** involves recognizing potential sources of harm, such as dangerous equipment, hazardous chemicals, or ergonomic issues.
- 2. Risk assessment** evaluates the likelihood and severity of harm resulting from these hazards and to prioritise them.
- 3. Risk control** focuses on implementing measures to mitigate or eliminate the risks, such as modifying equipment, introducing safer work practices or providing personal protective equipment (PPE).
- 4. Monitoring and reviewing** ensures that control measures remain effective and are continuously improved based on feedback, incidents, or changes in workplace conditions. This is further supported by an OSH information system that collects data on environmental exposure monitoring as well as health and safety such as occupational incidents, injuries and diseases, as well as fatalities.

Figure 1. Occupational safety and health (OSH) risk management cycle



Source: Safe Work Australia. 2018. Code of practice: How to manage work health and safety risks. In: *Safe Work Australia* [Cited 26 March 2025]. <https://www.safeworkaustralia.gov.au/doc/model-code-practice-how-manage-work-health-and-safety-risks>

How to perform Occupational Safety and Health Risk Assessment and Management

When developing an occupational safety and health management plan, it is important to identify the OSH issues of the specific workplace by performing an **OSH risk assessment**:

- Conducting informal inspections of the various work areas and tasks (Ngajilo, 2025)
- Discussing OSH concerns with workers and others who may frequent the workplace
- Consulting information on OSH industry standards and best practices – such as this Best Practice Guidance.

It is good practice to involve workers in all stages of this process – after all, they encounter the hazards on a continuous basis and therefore have intimate knowledge of the actual operational OSH risks.

This Guidance of Best OSH Practice approaches occupational safety and health in the inland aquaculture/fisheries sector from a risk-based perspective, noting that the process of risk management follows the simple 4-step process of the OSH Risk Management Cycle using a simple mnemonic:

SAFE (SEE, ASSESS, FIX, and EVALUATE)



Steps in the 'SAFE' OSH Risk Assessment and Management Process

Step 1: SEE the Hazard (Identify the Hazard)

A hazard is any situation, activity, procedure, piece of equipment/machinery or fish that may cause harm or injury to a person

Common hazards in inland aquaculture/fisheries and their adverse health effects are presented in the **Table 2**. It is evident while there are a number of hazards common to both sectors, some may be more common in one of the two sectors.

All tasks, equipment and substances should be examined. When identifying and listing hazards use:

- information from past incidents and workplace injuries
- information from workers as well as their families
- product literature and information from suppliers
- best industry practices
- sight, smell, touch and hearing senses
- close examination of areas or activities where children or visitors may be present

Step 2: *ASSESS the Risk*

Risk is the likelihood that an existing hazard may actually cause harm or injury, as well as the expected severity of that harm

Every inland aquaculture/fisheries operation has some element of risk – there are likely to be hazards that could potentially impact the OSH of workers. Operators need to ensure that workers can enjoy a healthy and safe workplace by adhering to OSH regulations, providing appropriate and adequate supervision, and ensuring that all workers have the necessary training and equipment to do their jobs in a manner that does not endanger their health and safety.

If a hazard has been identified, assess the risk by examining;

- The likelihood of the hazard resulting in injury or disease to workers or other persons - is it likely or unlikely to occur?
- The severity of the incident - could it cause death, serious injury/disease, or minor injury/disease?

To assess the risk of a hazard having the potential to harm someone, ask questions such as:

- How many workers or other persons come into contact with the hazard?
- How often is the contact?
- How seriously could someone be harmed?
- How quickly could a dangerous situation arise if something goes wrong with the equipment or the work process?

Step 3: *FIX the Problem* (Eliminate or Control the Hazard)

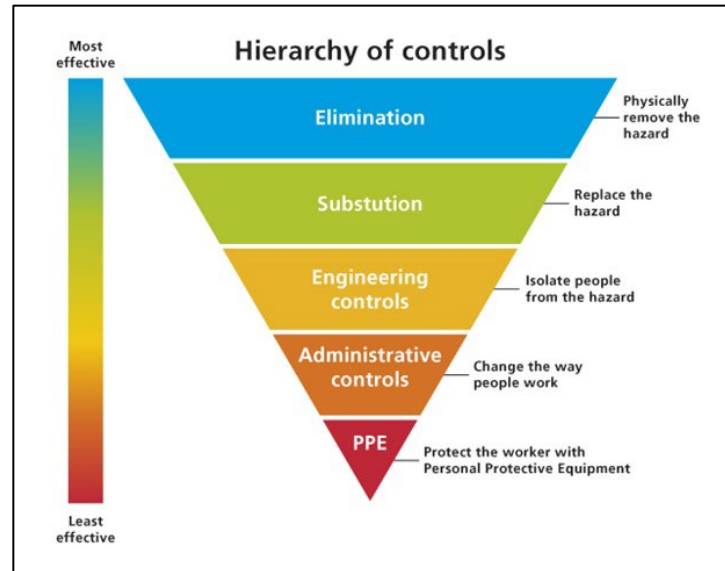
Use the hierarchy of control measures to eliminate or control the hazard

There are several ways to control a hazard. Pick the way(s) that is reasonable and practical for the actual context. Based on the hierarchy of control measures, which is aimed at prevention, these can be addressed in the following order:

1. **Eliminate** hazards posed by equipment, existing processes, and the environment if at all possible – this is the most desirable step! For example, get rid of a faulty machine.
2. **Substitute** with a safer measure by using a different chemical, machine, material or work practice that poses less risk to perform the same task. For example, use a chemical that is known to be less hazardous instead of a more dangerous one.
3. **Use engineering/design controls** when it is not possible to eliminate a hazard. Engineering controls include machinery guards and power take-off (PTO) shields. Design controls, such as ventilation, noise barriers or locked fences, that isolate the hazard from the worker.
4. **Use administrative controls** that focus on changing the way people work to reduce the duration, frequency or intensity of exposure to hazards. These can be done through having guidelines on job rotation, rest breaks, no smoking or eating when performing tasks, appropriate training and supervision to promote healthy and safe practices to minimise risk. For example, supervise new workers until they are competent to deal with potentially hazardous work.
5. **Protect workers** if other controls are inadequate by providing personal protective equipment (PPE). Use and provide proper clothes (e.g. overalls, gloves), eye and respiratory protection for handling dangerous chemicals or biohazards. Provide hard hats and boots for protection against safety hazards. Remember though, that use of PPE is not the most reliable due to general poor

compliance, so as far as possible, implement the controls suggested earlier in this list. However, in certain emergency situations where there is a temporary lapse in general control measures, respiratory PPE become crucial. Similarly, certain work processes will require use of safety gloves, cut-resistant gloves, or other specific types depending on the nature of the exposures.

Figure 2. Hierarchy of control measures



Source: National Institute for Occupational Safety and Health (NIOSH). 2015. Hierarchy of controls. In: *U.S. Centers for Disease Control and Prevention (CDC)*. [Cited 26 March 2025]. <https://www.cdc.gov/niosh/hierarchy-of-controls/about/index.html>

Step 4: Evaluate The Fix (Check that the control measure is working)

Evaluate whether your control measures are working on an ongoing basis

The final step in the process is to check whether the controls are actually working or whether workers are following them?

Ask:

- Have the chosen control measures been implemented as planned?
- Are the chosen control measures working?
- Are there any new problems (unintended consequences)?

To do this after implementing the “Fix”, one can do an inspection to monitor adherence. Secondly, direct measurement of a particular hazard (e.g. noise level) where appropriate. Regular health monitoring of workers using questionnaires (for symptoms), medical examination or special tests (e.g. audiogram). Finally, collection of information on the frequency of incidents, occupational injuries or diseases, and deaths will provide objective evidence of the effectiveness of control measures.

Consult the section on Best Practice Guidance for Managing Common Occupational Hazards for details. This guidance is based on various other codes of best practice and adapted for inland fisheries and aquaculture (ACWA, 2002; HSE, 2006; WCB PEI, 2020). ***While it should be noted that a Best Practice Guidance on OSH such as this one, cannot cover every possible activity in the inland aquaculture/fisheries sector, the common activities and associated hazards have been included. Each specific workplace will need to address additional activities where necessary.***

4. BEST PRACTICE GUIDANCE FOR MANAGING COMMON OCCUPATIONAL HAZARDS

4.1 Safety hazards

4.1.1 Slips, trips and falls

Slips, trips and falls from wet surfaces or heights, are amongst the most common causes of injuries in inland aquaculture/fisheries. These can result in muscle strain, broken limbs and contusions and death in the worst cases.

Best practice guidance for slips and trips

- Where possible, ensure that floor areas are non-slippery.
- Oil, ice or debris must be removed from walking surfaces immediately.
- Likewise for equipment, tools, obstacles and debris – remove from the immediate working area. Keep walkways clear at all times.
- If there are lines, cables or ropes that cross the walkway area, cover them where possible.
- Ensure that walkways and working areas are well lit. Measure light intensities in areas where appropriate. Replace broken light bulbs and faulty light switches.
- Workers should wear the appropriate anti-slip footwear for the task and actual working conditions.
- If walking surfaces are wet, icy, uneven, or cluttered, slow down the walking pace.
- It is important that objects being carried or pushed do not obstruct the view of any hazards in the walking path.
- Construct, and maintain all runways, ramps and platforms so that they safely support all normal loads.
- There should be a guardrail to hold onto, at least on one side of all runways, ramps or platforms.

Best practice guidance for: Ladder use

- Workers should only use a sturdy, secure ladder to access elevated or sub-level areas.
- All ladders must be kept in good working condition. If there are loose, broken or missing rungs, split side rails or other hazardous defects, remove the ladder from service.
- The ladder should always be secured at the top to prevent it from moving.
- Maintain 3 points of contact (e.g., two feet and one hand always on rungs or alternatively two feet plus a safety harness attached to a secured point).
- Always face the ladder when climbing. Do not overreach or jump off ladders.
- Use the “four-to-one-rule” for straight ladders – the ladder base is set one foot from a wall or building for every four feet in height.
- Place the ladder on a sturdy flat surface if the ground is soft or uneven.
- Stay away from power lines, and do not work at heights during adverse weather conditions or if ill, tired or taking medication that cause drowsiness.

4.1.2 Chainsaws

Chainsaws often used for winter harvesting and cutting holes in the ice. Sometimes they are used to clear wood debris around buildings. They are often associated with injuries if not used properly.

Best practice guidance

- Wear approved hearing protection, safety eyewear, safety pants, a hard hat and safety footwear.
- Operate a chainsaw that is in safe working order (equipped with a safety chain, chain brake,

anti-vibration mounts, throttle-control lock out, chain catcher and rear hand guard).

- Maintain the tool in good condition.
- Stop the chainsaw motor before refueling or performing maintenance work on it.
- Hold the chainsaw firmly in both hands while operating.
- Start the chainsaw away from any part of the body.
- Wear safety gloves or mitts while sharpening a chainsaw.

4.1.3 Equipment and Machines

Most machinery and equipment have the potential to cause injuries if not used in a safe manner. Common hazards associated with machinery/equipment which can potentially cause injury include:

- Gearing, cables, sprockets, chains, clutches, cams or fan blades
- Machine components that cut or grind, have shear points or crushing points
- Hot surfaces that can cause burns, fires or explosion
- Rotating power take-off (PTO) drive lines and other shafts, as well as all projections on rotating parts
- Fluids / gases under pressure can explode.

Best practice guidance

- Only use machinery/equipment according to instructions as intended by the manufacturer.
- Inspect power-operated machinery/equipment every day before being used.
- Always ensure safety before starting any machinery/equipment.
- Operators of machine/equipment must have unimpeded access to the shut off.
- Only competent persons are to operate the machine/equipment.
- Wait until machinery/equipment has stopped before climbing on / cleaning.
- Do not ride on the load bed / bucket of a vehicle while it is being used.
- Secure all tools and other potentially harmful equipment on vehicles – especially when workers are being transported.
- Wear close fitting clothing, cover the head or facial hair and avoid wearing dangling jewelry or rings when working around moving parts of machinery.
- Protect against burns caused by hot machine parts.
- There must be ample working space, ventilation and lighting when operating, adjusting or repairing machinery/equipment.
- Maintain powered mobile equipment, and keep their operation down to a minimum, especially when they are operated in closed areas for prolonged periods of time since they can emit dangerous chemical vapors.
- Back-up alarms on mobile equipment must be functioning properly.
- All machine guarding must be left in place and properly maintained.
- All moving parts of machinery should be guarded.
- Maintain all air and hydraulic lines, hoses and components in a safe working condition.
- Replace or repair all defective parts prior to use.
- Only lubricate machinery when it is at rest (or as directed by the manufacturer).
- Do not repair pressurized steam, air or hydraulic lines while machinery is in use.
- Keep children and bystanders away from those areas where tools, equipment or machinery are being operated or maintained.

4.1.4 Hoisting devices and conveyors

Hoisting systems or conveyors are often used in inland aquaculture/fisheries operations and can cause injuries. Common hazards associated with these items include:

- No lock-out procedures
- Untrained operators
- Machine guarding missing on moving parts (e.g., pinch points, wrap points)
- Workers riding on the load or the equipment
- Lack of personal protective equipment (PPE)
- Poorly maintained or damaged equipment.

Best practice guidance for: Operators and crew

- Workers must be trained before operating equipment or mechanical systems.
- Stand clear of the load being lifted. If the load is heavy, rather push than pull the load by hand. Do not try and help by lifting the load.
- Always lift from directly over the load.
- Never pass a load over the head of other workers.
- Do not stand under the object being lifted.
- Start moving the load only when everyone is clear of the load – slowly and smoothly!
- Do not ride on the lifting system or the load that is being lifted.
- Do not raise the load any higher than necessary, nor should it be left suspended / unattended.
- All workers involved in the lift must know the communication signals for stop, go, lift the load, etc.
- Only those involved in the lifting operation should be near the lifting system (or conveyor) while it is in operation.
- Before working on a conveyor, the equipment must be stopped and locked out.
- Workers must not stand on the supporting frames of an open conveyor while loading, unloading or clearing blockages.
- Remember to wear gloves if needed, tie up long hair, and avoid wearing loose clothing or jewelry when working near a hoist, boom or conveyor.

Best practice guidance for: Equipment

- Inspect lifting systems regularly. Worn or damaged components must be replaced before using.
- Ensure all moving parts (e.g., pinch points, rollers) are guarded.
- If used outdoors or in damp conditions, ensure all electrical components are waterproofed.
- Ensure that the safe load limit is posted in a prominent position. This limit should not be exceeded.
- Always position lifting devices like hooks, rope loops correctly before lifting the load.
- Repair lifting systems according to the manufacturer's guidelines. Remember to switch off and lock out first!
- All workers MUST know the location of the emergency shutoff and how to use it.

Best practice guidance for: Rope safety

- Ropes used must always be inspected first before use. Look for damage or chafing.
- Never exceed the safe working load of the rope.
- Make sure there are no sudden strains or jerks on the rope – it can snap.

- Keep the rope away from chemicals, direct sunlight or dampness – these can weaken it.

Best practice guidance for: Blocks, sheaves and other fittings

- These parts of the lifting gear should be regularly inspected, lubricated and maintained.
- Ensure that sheaves are properly guarded.
- Regular inspection is needed for all bushings, nuts, bolts, etc. Check for signs of wear and replace where necessary.

4.1.5 Hydraulics

Hydraulic systems are very useful, but if they are not properly maintained or adjusted they can cause serious injury or death. In addition, the hydraulic fluid contained within the system may be toxic and can cause severe burns. Two of the main routes of entry into the body are the eyes and skin, which can suffer adverse health effects.

Only experienced people should perform maintenance tasks.

Best practice guidance

- Always follow safety procedures in the manufacturer's manual.
- Only properly trained personnel are to operate equipment or mechanical systems.
- Be careful when working near raised hydraulic cylinders.
- Ensure defective parts on hydraulic systems are replaced immediately. A planned maintenance system can be very useful.
- Remember: do not check for high pressure leaking hydraulic fluid with your hand – use a piece of cardboard or wood over the suspected area. In instances of a hydraulic skin puncture, seek medical attention immediately.
- Always deactivate any hydraulic powered equipment before servicing or repairs:
 - lower it to the ground or block it mechanically,
 - shut off the engine that powers the hydraulic pump,
 - release hydraulic pressure (i.e., move the hydraulic lever back and forth several times),
 - let the system cool down, and
 - follow the manufacturer's servicing procedure instructions.
 - Make sure any spilled hydraulic fluid is cleaned up immediately.
- Always dispose of used hydraulic fluids in a safe and environmentally-friendly way.

4.1.6 Hand and power tools

Workers need to understand the hazards associated with the equipment and tools found on the worksite. Incidents are often directly linked to:

- tools being used for the wrong task
- cluttered, poorly lit and unventilated areas
- faulty electrical cords
- tools and equipment without the required guards

Power tools can cause severe injury. Identify potential hazards in all workspaces. Thereafter, develop safe working procedures for your work area and tasks. All workers must be informed of these procedures and be able to carry out the tasks safely.

Best practice guidance

- Make sure workers have been properly trained in the safe use of all tools.
- Tools must be kept in good condition. Replace or repair defective tools before using.
- If the casing on tools is broken or damaged, do not use!
- Ensure all damaged cords or plugs on power tools are properly repaired or replaced.
- Power tools must be shut off, and the plug removed from the outlet before making adjustments.
- Do not use makeshift extension lights.
- Ensure that all extension lights have a guard around the globe and an insulated handle.
- All cutting tools must be kept sharp, and sharp edges covered when not in use.
- Make sure that sharp tools (e.g., saws, knives) used on work benches are pointed away from the work area. Do not allow handles to hang over the edge of the bench top.
- Always wear safety glasses or goggles and well-fitting gloves should the task require it.
- Tools must only be used for the purpose they are intended. Do not apply excessive force or pressure on tools.
- Cut away from yourself when using cutting tools.
- When climbing a ladder, be careful that the tools being carried do not interfere with the use of both hands on a ladder.
- Be careful with sharp tools. Do not throw or carry them in your pocket.
- Tools must be looked after at all times. Maintain and keep them clean and dry.
- After using, store tools properly.
- Remember housekeeping! Always keep work areas clean and tidy.

4.1.7 Electrical

Electricity, while essential to most operations, has the potential to cause sudden and serious harm, including death. Proper care and attention should be given to the maintenance and use of electrical units.

Best practice guidance

- Electrical safety rules must always be followed.
- During repair or maintenance, main switches must be locked out, locks and tags placed in them before working on power circuits. Make absolutely sure that no one else can use them while one is working on the circuit.
- Ensure that periodic electrical inspections are conducted by qualified persons.
- Only qualified individuals should be allowed to work on electrical repairs.
- Ensure that all wiring, equipment, leads and plugs are kept in good repair.
- Wiring installation must not be overloaded.
- Always consider every circuit to be live – and test appropriately before repairs/maintenance.
- Ensure that extension cords and devices are properly grounded.
- Only use the correctly rated fuse or circuit breaker when replacing them.
- All power tools (portable or bench-mounted) should be properly grounded.
- Avoid using electrical equipment in wet weather or working conditions where possible. Furthermore, only properly ground electrical equipment must be used in wet areas (use a ground-fault circuit interruption outlet).
- Always use weather-proof outlets and fittings, especially in areas exposed to the elements or where the presence of moisture could routinely pose a hazard.

Best practice guidance: Overhead power lines

When operating equipment in any location where overhead lines are present, the following points are important:

- Always be aware of overhead power lines - maintain minimum approach distances.
- Be aware of the operating height of equipment. Note extensions that add overall height.
- Lower the equipment prior to moving under power lines.
- Remember that factors such as snow build up on the ground, snow and ice loading on overhead lines, as well as warm and humid weather or other activities, may cause a reduction in the distance between the lines and the top of any equipment operating below.
- If equipment comes into contact with electrical lines, remain calm and have someone call the local electric utility. Warn others to remain clear. Do not step off the equipment while it is energized – wait for everything to be switched off first.

4.1.8 Fires

Fires and explosions can cause serious injury or death due to the inhalation of toxic smoke and vapours or serious burns. There are several possible causes of fires in inland aquaculture/fisheries operations and vessels. These include improper storage of fuels and / or refuelling activities, faulty wiring and overloaded outlets, leaking propane tanks, overheating equipment, smoking, and welding operations. Even the improper storage of oily rags can start a fire.

Best practice guidance

- All workers should know what to do in a fire emergency.
- Electrical systems should be inspected regularly – especially if there are problems with flickering lights, disruptions of power, etc. Circuits must not be overloaded.
- Ensure that all extension cords are kept in a good condition.
- Use wire cages to protect light bulbs and heat lamps in hazardous situations.
- Place fire extinguishers at the right place appropriate to the fire hazard.
- Keep fire extinguishers fully charged and in good condition.

Best practice guidance: Storage of flammables/combustibles

- Flammable and combustible materials must be stored away from oxygen tanks and sources of heat (e.g., open flames, engines, equipment, daylight).
- Fuel and flammable material must be stored in good condition, labeled containers.
- Do not let flammable/combustible materials block/obstruct exits.
- Only keep necessary amounts of flammable or combustible materials on board a boat – just enough for daily maintenance and operational needs.
- All work requiring the use of flammable materials must be undertaken in a well-ventilated area. Keep area clear of anything that could produce a spark.
- Never refuel a running or hot pump/motor.
- Have a “no smoking” policy in place in hazardous areas.

Best practice guidance: Boats/vessels

- Ensure that each vessel has the appropriate fire extinguisher(s) on board.
- Propane appliances and cylinders must be properly secured, protected and be stored in well-ventilated areas. All valves must be work properly, and all connections tight.
- Bilges must be kept clean and free from oil and fuel. Bilge pump must work well.
- Enclosed engine spaces must have proper ventilation.

- Batteries must be stored correctly, and maintained properly.
- Electrical systems on the boat must be correctly, and safely installed and maintained.
- Adjust and lubricate machinery to minimize friction.

4.1.9 Sharps

Workers in the inland aquaculture/fisheries industry are often exposed to the hazards from objects that can penetrate or puncture the skin (like knives, needles, etc). It is important that precautions are taken to reduce the risk of injury and associated adverse health effects associated with these tools.

Common hazards

The types of hazards that are common when working with sharps include:

- The use of needles for inoculating fish: accidental punctures when using these (this is the most common cause of needlestick injuries)
- The use of knives and screwdrivers: limb injuries caused by accidental stabbing or cutting
- The storage of these pieces of equipment: limb injuries caused by improper storage or disposal

Best practice guidance

- Always be very careful when handling all sharps (needles, knives, screwdrivers, box-knives, razors, etc.).
- Housekeeping and organizing the work area will help – especially if sharps are located in a safe manner for the work to be done.
- All workers using needles should be properly trained to handle the needles and perform the required tasks (e.g., inoculating fish).
- Consider mechanising the inoculation of fish should resources permit.
- Use appropriate gloves to minimize hand injuries.
- It is important that needles are kept capped until ready for use.
- Never insert your hands into a sharps container unless you are absolutely sure it is empty.
- All sharps must be discarded in a rigid, wide-mouthed, leak-and-puncture-proof container. This container must have a lid that can be tightly closed. The container and its contents must be clearly identified as containing sharps.
- The sharps disposal container must be placed in a convenient section of the work area and close to where needles are to be used.
- Ensure the location is clearly signposted.
- It is good practice to replace the sharps disposal container before it is full.
- Sharps disposal containers must be safely and properly disposed of.

4.1.10 Finfish

Common hazards

There are numerous hazards when working with finfish that may cause injuries and toxic reactions, including:

- Limbs, clothing, hair or jewelry can get caught in machinery / equipment
- Trips and falls when walking over rough terrain or climbing ladders while carrying heavy loads of fish
- Transporting, loading and handling fish in large tanks
- Crush injuries when caught between the fish tank and another structure
- Compressed gasses (e.g. oxygen) in the transport tank

- Falling from heights such as platforms while working with fish
- Possibility of injury when handling large, powerful fish (e.g. salmon)
- Some fish are poisonous since they have sharp fins through which they deliver venom
- Decomposing fish in tanks can cause high levels of toxic gases such as hydrogen sulphide

Best practice guidance

- Workers must be trained on the proper techniques and precautions for performing tasks associated with fish handling.
- There must be clean walkways so that workers have good footing when treating and handling fish.
- There should be minimal lifting required to transfer the fish into anesthetic baths or tanks.
- Fish should be fully anaesthetized before removing them from the anesthetic bath.
- Use ergonomic principles to ensure worker comfort and safety when working with fish or for any work requiring heavy lifting or prolonged repetitive motions.
- Ensure good footing at all times when walking in tanks - especially if work involving fish is completed before, or without, completely draining the tank.
- Do not jump over the top edge of the tank or drop into the tank – use a ladder that hooks over the tank edge.
- Always ensure that there are railings surrounding ground level tanks, as well as appropriate handrails along all walkways, around the edge of cages or over tanks.
- Walkways must be kept clear and free of debris (including ice and snow).
- Always wash hands regularly – especially after cleaning tanks and handling fish.
- Appropriate fish-handling gloves should be used to minimize hand injuries.
- After working on any fish or after using chemicals (e.g., inoculants), make sure that work surfaces are cleaned and disinfected at the end of the day.
- Ensure that all workers that work with a hazardous chemical, are trained in the procedures for the safe use, handling, storage, and disposal of the product.
- Use special air-purifying respirators with specialized canisters/cartridges for hydrogen sulphide when entering tanks that contain dead and decomposing fish.

4.1.11 Land transportation

Inland aquaculture/fisheries workers use many modes of transportation to carry out their daily tasks, which may cause accident-related injuries. Each of these has specific health and safety requirements. A good place to start is to review the operating manual and manufacturer's information before any vehicle is used.

Best practice guidance

- All operators of work vehicles must be properly trained.
- Vehicle operators and/or passengers must wear the appropriate personal protective equipment.
- Ensure that all vehicles are in proper working order before being used, and are all regularly inspected and maintained.
- Always perform pre-ride inspections and periodic maintenance of the vehicles (as outlined in the owner's manual).

Best practice guidance for: Vehicles

- Travel speed must be adjusted to the road conditions and activities being undertaken.
- Ensure that all vehicles have the proper lights needed for working in dark or inclement weather

conditions

- Passengers should not be transported, nor stand on tractors, equipment or trailer while underway.
- All guards must be kept in place on all equipment.
- Implements must be attached to the appropriate hitch provided by the manufacturer.
- Power take-off (PTO) tractors must be switched off before hitching implements.
- Install rollover protection structures (ROPS) on tractors to prevent tractor overturn.

Best practice guidance for: Trailers and wagons

- The hitch and safety chains must be properly connected when a trailer is being towed.
- The trailer load must be firmly secured, and properly loaded for stability.
- Ensure that items such as tyres and brakes on both the towing vehicle and trailer are in good condition and that brakes are always sufficiently strong for the load.
- Drivers must be experienced and properly trained in operating the towing vehicle and attached implements.

Best practice guidance for: All-terrain vehicles (ATVs)

- When using an ATV, wear the appropriate PPE (helmet, eye protection, boots, long sleeved shirt or jacket, long pants, gloves).
- Always operate the ATV at a speed appropriate for the conditions.

4.1.12 Boats and boating

Boats are often used in inland aquaculture/fisheries farms, but their use can pose a considerable hazard to those that work on them. The size and type of boat will vary according to circumstances and needs, but regardless of the size of the boat or operation, the associated hazards causing injuries or disease, often remain the same.

Common hazards

- Navigation, collision and grounding
- Fire or explosion
- Engine or other mechanical difficulties
- Flooding and capsizing (especially with shifting loads)
- Persons falling overboard and drowning
- Weather-related conditions such as sunburn, sunstroke, hypothermia or lightning
- Entanglement in gear or hauling equipment
- Cuts from knives, lines, tongs etc.
- Hearing damage due to loud noise

Best practice guidance

Basic components for vessel and crew safety are:

- **Vessel:** must be stable, fit-for-purpose, and seaworthy
 - Boats must be safe and suitable for the operation for which they are to be used.
 - Regular maintenance necessary – daily pre-trip inspection of the fuel, oil and/or other fluid levels.
 - Machinery/equipment must be maintained in a safe condition.
 - Maintain stability – keep weight down low.

- Carry a tool kit, spare parts (e.g., fuses, impellers, bulbs, duct tape, spark plugs)
- Have the necessary means to stop small leaks so that repairs can be carried out in the event of an emergency.
- Fuel containers and compressed gas tanks must be stored properly.
- Equipment and gear must be stored properly.
- Slippery areas must be cleaned immediately.
- Working areas must be kept clear and uncluttered.
- Heavy equipment or gear must be secured to avoid them shifting.
- Keep a first aid kit that is stored in a dry place. Replace used and outdated contents.
- Keep accessways and ladders in good condition.
- **Crew:** must be properly trained, rested and competent
 - Never stand on any lines, wires, cables or ropes that are under tension.
 - Each crewmember must know how to use the appropriate personal flotation devices. When the water is extremely cold and the risk of falling in is high, a floater suit or other survival gear should be worn at the outset.
 - Appropriate clothing for the season and task must be worn at all times – no loose-fitting clothing or jewelry should be worn near moving equipment, lines, etc.
 - Appropriate gloves are to be worn for hand protection when required for certain work processes.
 - Boots/shoes must have good traction.
 - Wear safety glasses or goggles when the activity requires eye protection (e.g., cleaning with chemicals, hauling lines).
 - When reaching over the side/stern, always have one hand holding a railing or another piece of the boat.
 - Only use the provided means of boarding/exiting the boat (e.g., ladder).
 - Avoid getting jammed between the boat and the dock or wharf when boarding or exiting the boat or when tying off.
 - Always keep a sharp knife available to cut jammed lines. Store the knife safely.
- **Lifesaving/rescuing equipment:** must be in place and ready for use
 - Each vessel must be fitted with the appropriate safety equipment.
- **Equipment and operating procedures:** must be fit for purpose.
- **Fire protection:** provide fire protection equipment and training to promote fire safety awareness and practice regular drills.
- **Maintenance:** must be based on standards and follow best practice.
- **Communications system:** must be in place, understood and used.
- **Loading/unloading the Vessel:** Safe work practice at all times
 - Ensure the safety of persons loading or unloading the vessel.
 - Ensure safe and even loading of equipment and product/s.
 - Always consider weather conditions to prevent overloading or capsizing.
 - Keep load and equipment as low as possible and keep it secured to prevent cargo shifting or affecting the boat's stability and buoyancy.
- **Hauling equipment:** must be kept in good working order
 - Inspect equipment and gear often, and replace anything that is worn, broken, has excessive wear and tear, or is otherwise unsuitable/unsafe.

- Stand clear of the hauler where possible.
- Hands must be kept clear of the line at all times.
- Stay clear of the moving gear (e.g., line, cages, buoys). Never stand under a load or in areas where overhead equipment may swing or drop and cause serious injury.
- **Safe navigation practices:** must be understood and adhered to by all crew
 - Avoid high-speed operation near shorelines, other boats or structures.
 - A safe boating speed will depend on the visibility conditions, wind/water conditions and currents, maneuverability of the boat, boat traffic in the area, and the proximity of any navigational hazards.
 - Ensure a member of the crew has knowledge of shipping routes, markers and hazards as well as the prevailing tides, currents, weather patterns to ensure the safe operation of the boat.
 - Ensure a member of the crew knows the rules of navigation.
 - Ensure that all navigational safety equipment (sound-signaling devices, radar reflectors, navigation lights) are in good working condition and appropriately located onboard.
- **Weather conditions:** need to be assessed before operations
 - Where appropriate, only undertake operations when the long-range forecast is suitable to ensure the safe completion of the task.
 - Always watch out for signs of approaching bad weather (e.g., choppy seas, cloud bank in the distance, a rise in the humidity, and any sudden changes in the atmospheric pressure). Monitor weather updates until returning to port.
 - If the weather takes a sudden turn, know what to do in order to protect the crew (e.g., nearest shelter, return to port procedures).
 - In an electrical storm, lower the hauler/boom. Avoid contact with metal surfaces and electrical equipment. Take shelter in the cabin (if available).

Rescue procedures for “man-overboard” situations

Falling overboard can result in death very quickly as a result of drowning or cold-water exposure. Ensure that there are procedures in place for recovering an overboard crew member.

Best practice guidance

- Immediately throw a life jacket, life ring, or a bright colored floating object into the water to help the person, and to guide the vessel back to the overboard spot.
- Inform the skipper immediately. Verbally, direct the vessel operator to the overboard crew member.
- ALWAYS keep the overboard person in sight.
- The vessel must carefully maneuver alongside the overboard person so as not to cause further injury. Rescuers must be careful not to end up in the water themselves.
- Immediately apply first aid / resuscitation as required. Carefully watch the rescued person in case there are complications.

4.2 Physical hazards

4.2.1 Diving

Diving is often an important part of inland aquaculture/fisheries operations. Divers are required to inspect lines, anchors and the general condition of the seafood operations, as well as assisting with other plant operations. Divers are required to be knowledgeable about the type of diving they are about to carry out, the conditions at the dive site and the tasks they are about to perform. All diving

operations must be well planned and be carried out by experienced divers with the proper knowledge, training and experience to carry out their assigned work. Hazards associated with this activity include entanglement with nets and underwater entrapment and drowning.

Best practice guidance

- All divers should be qualified in First Aid.
- Divers should be instructed in the work procedures they will be required to carry out.
- All divers must be able to swim well, and where possible, always dive with a ‘buddy’.
- Divers must only dive if they are fit to do so (e.g., not ill, fatigued, impaired, injured). They must be assessed by a doctor regarding medical treatment, medical conditions or medication being taken, that may make them unfit to dive.
- Divers must always consider weather and water conditions in the dive area – and suspend activity if it is hazardous or likely to become hazardous.
- All diving operations must follow the applicable OSH regulations.
- Divers must remain within the posted dive site.
- Where necessary, divers must be provided with a lamp or other means of illumination during periods of darkness.
- Snorkel divers must always use a buddy system, and should be equipped with a whistle, weight belts (if required) with quick release closures, as well as thermal protection.
- Vessel propellers should be caged or props on dive boats must be disengaged and locked out while being used as a dive base.
- Divers must carry an audio or visual signalling device when conducting a dive in open water without a lifeline.
- Divers must be protected from all hazards at or near the dive sight.

A written dive plan, specific to the particular dive site, must be prepared before a dive is conducted. This dive plan must include a description of the tasks to be performed at the site, including work procedures for each type of dive and the diving equipment to be used, how long the dive will be, and the decompression tables and procedures to be used. In addition, the procedures to identify and address OSH issues at the dive site must be clear, with a list of all hazards in the immediate area of the dive site, and instructions for getting medical assistance and the possible evacuation of an ill or injured diver from the dive site, as well as any other emergency responses procedures needed. These could include loss of communication, diving equipment malfunction, hazardous weather or water conditions, aborting a dive or responding to any difficulties in keeping the dive base stationary.

4.2.2 Weather

The weather can pose many hazards, especially in occupations like inland aquaculture/fisheries, where much of the work is done outside. This ranges from exposure to the heat and sun during the hot summer months, and to the cold and wind during the winter months. Stormy or inclement weather on the water can also put workers at risk. Working in very cold and icy weather is particularly hazardous, and care must be taken to ensure the OSH of workers.

Common hazards outdoors include working in the heat and sun. Excessive heat exposure can cause dehydration, heat exhaustion and fainting as well as mental confusion as a result of heatstroke. Furthermore, exposure to solar ultraviolet radiation can cause severe sunburn and in the long-term, skin cancer.

On the other end of the temperature scale, prolonged exposure to cold weather, particularly accompanied by water-soaked clothing or direct immersion after falling overboard, can result in dangerous overcooling of the body. Here, hypothermia can affect a person’s mental and muscle function and ultimately result in death.

It is important to remember that victims of heat stroke or hypothermia are often unable to notice the symptoms of the illness. So, it is important that their co-workers are able to identify the symptoms, provide emergency assistance and seek urgent medical help.

Best practice guidance: Working in hot weather conditions

- Schedule working time during cooler parts of the day, if possible.
- Workers must be provided with ample drinking water, and short rest breaks in a cool area.
- Use light, breathable clothing to enhance comfort and reduce heat stress and exhaustion
- Workers should wear appropriate additional protection such as sunscreen and a hat. Wearing sunscreen and a hat keeps the direct sunlight off the head and face area.
- Workers should wear sunglasses to protect their eyes from direct sunlight or glare of the water.
- Always drink lots of fluids, especially water. The amount of diuretics (coffee, cooldrinks, tea) consumed should be reduced since they increase fluid loss.

Best practice guidance: Working in cold weather conditions

- Where possible, restrict the time that workers are required to work in extreme temperatures (e.g., take frequent short breaks instead of infrequent long breaks).
- The likelihood of hypothermia can be reduced by wearing warm, protective clothing and keeping the head and extremities (hands, feet, arms, legs) covered.
- Remember, heat is lost more rapidly through the head and neck, sides of the chest (including the armpits) and the groin area.
- Learn the appropriate methods to stay warm, especially if someone falls into the water. This includes taking deep and controlled breaths while still immersed in the cold water. Once out of the water, dress immediately starting with the top half of the body, dry the body and remove wet clothing, pat the body vigorously, and layer up with warm clothing.

4.2.3 Noise

Exposure to high noise levels in inland aquaculture/fisheries may occur when working with vehicles, heavy equipment (like tractors and cranes), vibrating feeders, generators, pumps, and power tools. In addition, constant movement of water required during intensive aquaculture (e.g. hatchery), makes occupational noise exposure a health risk in these operations. If workers are having to shout while speaking to other co-workers, this suggest noise levels may be higher than 85 decibels, capable of causing noise-induced hearing loss if exposed over a long period of time.

Best practice guidance

- Measure the noise levels to confirm the levels where this is identified as a potential hazard.
- Add signage where these noise zones are confirmed.
- Reduce the noise levels according to the hierarchy of control measures – fix machinery, enclose the machine if possible, install noise barriers (noise covers on tanks/stand pipes).
- Provide appropriate PPE – ear plugs/muffs when the noise levels cannot be reduced effectively or while using noisy mobile equipment or short duration noisy activities.
- Workers exposed to high noise levels on a long term basis should have their hearing tested on a regular basis.

4.3 Chemical hazards

4.3.1 Chemicals

Proper handling, transport and storage of chemicals are critical for maintaining the health and safety of anyone that comes into contact with these agents. Chemicals can pose a hazard to workers while using

them as well as to the environment. In aquaculture various liquid chemicals are used as additives to the work process. These include formaldehyde, antibiotics, disinfectants, pesticides, delousing agents, sulphites, fertiliser, Ph adjustment, and fish hormones. Other chemicals used in inland aquaculture/fisheries operations, are associated with the routine use of vehicles, vessels and equipment. There are also additional risks from materials used for cleaning fouling from gear (e.g., hydrated lime). These chemicals can affect different parts of the body and can cause acute irritant or allergic symptoms of the eyes, nose, airways and skin. Long-term exposure over many years can cause chronic health problems that can affect different body systems.

Best practice guidance: Use of chemicals

- Only use chemicals for their intended purpose, and under the prescribed conditions.
- Workers must be trained in the safe handling of the chemicals they use.
- All workers must familiarize themselves with adverse health effects associated with the specific chemicals they handle during the work process and what precautionary measures to follow. Safety data sheets (SDS) contain this information.
- Use appropriate personal protective equipment (PPE) at all times.
- Install extraction ventilation and use of timer pumps for reducing exposure to formalin in hatcheries.
- A petrol or diesel generator used in a confined space can cause carbon monoxide poisoning and loss of consciousness – make sure that adequate ventilation is present or relocate it outside.

Best practice guidance: Transporting of chemicals

- Ensure that hazardous chemicals are secured during transport. Do not leave chemicals unattended unless the container is locked up.
- Keep containers separately from workers if both are being transported together on a truck.
- Keep proper records of all chemicals transported.

Best practice guidance: Storage of chemicals

- Chemicals must be properly stored in a well-lit and well-ventilated area.
- Highly reactive chemicals should be stored separately (there is always the risk of explosions).
- Keep chemicals stored in the original containers with the labels intact.
- Do not store chemicals in food or beverage containers.
- Warning signs and emergency numbers must be posted in all storage areas.
- Safety Data Sheets (SDS) must be kept on file for all chemicals used.
- Absorbent materials must be available close by to clean up spills.
- Only keep reasonable quantities of fuel, oil and chemicals stored on site – do not stockpile chemicals.

Best practice guidance: Disposal of chemical containers:

- Empty containers to remove all traces of the chemical and clean them thoroughly.
- Dispose containers safely and return all empties to point of sale if possible (see specific SDS for details).

4.3.2 Welding, cutting and soldering

The use of gas and arc welding are common practices on many inland aquaculture/fisheries operations, especially when building or repairing of machinery occurs. Appropriate training should be given to workers given this task and all precautions followed in order to ensure the health and safety of the welder and any bystanders.

Typical hazards that welders face when welding include inhaling toxic fumes produced by the welding process, which can result in chronic chest problems including occupational asthma and lung cancer; eye damage (from the ultraviolet and infrared radiation) if no eye protection is used resulting in blood shot eyes, eye pain, light sensitivity, blurred vision and cataract formation; burns to the skin and eyes from flying hot metal bits. Other non-chemical hazards include fire/explosions from working with/around flammable liquids and compressed gases, excessive noise levels, electrical shock and ergonomic hazards due to repetitive and awkward work postures.

Best practice guidance: Personal protective equipment

- Always wear the appropriate eye, face, hand/arm, head and body protection (leather gloves, leather apron, gauntlets, safety glasses with side shields, welder's helmet or welder's goggles) as well as safety boots.
- N95 welding particulate respirators should be used, especially if working in a poorly ventilated area.

Best practice guidance: Facilities

- There must be adequate ventilation to remove any toxic fumes.
- Remove combustible materials from the working area and provide adequate fire extinguishing equipment where heat is used.

Best practice guidance: Welding equipment

- All welding / cutting equipment must be kept in good condition.
- Damaged equipment must be repaired to specifications by a competent person.
- Stop work immediately if a gas leak develops, and repair the damage.
- Store, and use compressed gas cylinders in an upright position. Secure correctly and do not drop cylinders or allow them to be subjected to impact.
- Cylinders containing flammable compressed gas must be separated by an approved fire-resistant partition from welding / cutting operations.
- Protective caps on gas cylinders must be used when they are being moved or not in use.
- Keep cylinders at a safe distance from all operations that produce flames, sparks, heat.
- Ensure all hose lines are properly marked, and fit reverse flow check valves as needed.
- Devices for holding the cylinders must allow for quick removal in case of fire.
- Only put down a welding or cutting torch when the gases have been completely shut off.
- Welding or cutting torches must not come into contact with the gas cylinder.
- Do not weld in wet conditions.

4.4 Biological hazards

Biological hazards are organic substances that pose a threat to the health. Biological hazards include micro-organisms (bacteria/viruses/mould), toxins (from poisonous fish/insects), bio-active substances (e.g. endotoxin) as well as fish protein/fish feed (containing allergens). Workers are exposed to these hazards while working in ponds and waste water containing discharging effluent (containing fish feed, metabolic waste or remains of dead fish, drugs and chemicals) as well as feeding, handling, slaughtering and filleting fish. Workers can acquire infections from contaminated ponds and effluent, toxic reactions from poisonous fish or insects, and allergic reactions (dermatitis, asthma) due to contact with fish proteins and organic fish-feed.

Best practice guidance

- Prevent direct contact with effluent in fish ponds by using appropriate protective clothing – but prevent seepage through protective clothing.

- Workers with cuts and bruises must be treated to prevent skin infections.
- Use appropriate gloves when handling fish with poisonous spikes.
- Use pellets for fish feed since they contain less feed dust that can be inhaled. Automated fish feeding further reduces exposure to dust.
- Ensure adequate ventilation and use appropriate respirators to prevent inhalation of bioaerosols.
- Use safe procedures to repair and clean up mouldy, damp structures.

4.5 Ergonomic hazards

Ergonomic hazards are common in inland aquaculture/fisheries. They refer to any physical condition or factor found in the workplace that can cause injuries or adverse health effects in workers. These conditions include carrying heavy loads, exerting excessive force, working with vibrating tools, performing repetitive motions and working in awkward positions for particular tasks. When these activities occur over a long period, they cause musculoskeletal problems such as low back, neck, shoulder and wrist pain including carpal tunnel syndrome. These hazards can be addressed through re-designing the workplace, including mechanisation or reorganizing certain tasks.

Best practice guidance

- Provide mechanical or powered aids to assist with carrying or moving materials and articles, where this is possible. If not practical, ensure adequate help is available to lift heavy or bulky objects.
- Provide floor mats in areas where prolonged standing is required.
- Choose power tools and equipment with features designed to control or limit vibration – wear appropriate gloves.
- Where possible, fit the work area to the worker's body size and shape in order to decrease the effort required to complete the task.
- Avoid working in one position for long periods of time.
- Reduce the speed of having to perform certain repetitive tasks.

4.6 Psychosocial hazards

Psychosocial hazards are factors in the work environment that can cause stress, strain, or interpersonal problems for the worker. Working alone or in remote settings in inland aquaculture/fisheries, is one such aspect that can be common in certain sectors. Working alone means that there is social isolation and loneliness. Furthermore, assistance is not readily available to the worker in the event of injury, illness, or an emergency. The OSH risk will depend on the location, type of work, and the consequences of an emergency incident or injury.

Best practice guidance: Working alone

- Use practical written procedures to ensure the health and safety of the worker working alone:
 - the name and contact details of the employer
 - identification of the possible OSH risks associated with the work assigned to each worker
 - the steps to be followed to minimize the risks identified
 - details of the means by which a worker can secure assistance in the event of injury or other dangerous incident

- Specify details of the:
 - time intervals for checking on the worker
 - person responsible for contacting the worker and recording the results of the contact
 - process to be followed if the worker cannot be contacted including provisions for an emergency rescue
- Implement a training program for each worker who is working alone and their supervisor.
- Facilitate broader social engagement on a regular basis, where possible.

5. CROSS-CUTTING OCCUPATIONAL SAFETY AND HEALTH (OSH) IN INLAND AQUACULTURE/FISHERIES

5.1 New and young workers

New and/or young workers often lack specific knowledge about occupational safety and health issues (OSH) at the workplace. These workers particularly, should be given hands-on instruction and supervision, and thereafter monitored periodically to ensure that they are doing the job properly and following all OSH procedures and protocols.

General precautions

- Ensure that new and young workers are familiarized with all hazards in the workplace.
- Use past incidents or “close calls” to illustrate the hazards and their adverse consequences.
- Emphasize that only trained workers are allowed to carry out certain tasks.
- Review safe working practices with new and young workers.
- Ensure that all new and young workers are properly trained.
- Ensure that new and young workers use the required personal protective equipment (PPE) for each task.
- Encourage reporting of hazardous situations, chemicals or equipment to their supervisor immediately.

Best practice guidance

- Supervisors of new and young workers must:
 - ensure that they have written work procedures where necessary
 - point out and explain any hazard, safe work procedure and PPE requirements for each task
 - provide a copy of safe work procedures for the task
 - ensure that the worker has enough time to become familiar with the workplace / task to be completed
 - demonstrate each step of the job procedure and detail all health and safety precautions, and have the worker perform the task under supervision of a trained person before allowing them to attempt the work by themselves
 - keep written records of the training workers, noting:
 - who provided the training,
 - who received training,
 - when the training occurred,
 - what training occurred.
 - ensure that workers are aware of his/her legal rights and responsibilities
 - correct any worker’s unsafe work habits and reinforce correct behavior
 - indicate when it is not appropriate to be smoking or eating on the job

5.2 Workplace housekeeping

Effective workplace housekeeping – on land or on the water – is in many ways the most important safety task that can be undertaken. Housekeeping includes keeping the work area neat and tidy, reducing slip and trip hazards in the work area, removing fire hazards from the work area and ensuring general maintenance.

Good housekeeping practice should include proper inspection, maintenance, upkeep and repair of tools, equipment, machines and processes.

Common hazards

Poor housekeeping contributes to workplace injuries from:

- being hit by falling objects
- tripping over objects on the floor, stairs and platforms
- slipping on wet, greasy, dirty or icy surfaces
- walking into projecting items and stacked materials
- cutting, puncturing or piercing the skin by projecting nails, wire, and other sharp objects

Best practice guidance

The general rule can be summarized as **“A place for everything, and everything in its place”**

- There must be proper storage facilities for all flammable, combustible, toxic and other hazardous materials.
- Materials must be stored in a safe and orderly manner. Ensure that that all stacking is stable.
- Collect waste regularly and dispose it appropriately.
- Clean spills immediately.
- Replace or fix broken or damaged items as soon as possible.
- Ensure that work / storage areas are well lit. Replace lights when needed.
- Tools should all have a designated storage area. Store after use. Do not place any tool or object in a place where it may pose a hazard.
- Where possible, use mechanical aides for carrying materials and supplies.

5.3 Washrooms

Proper **washing** and toilet facilities in outdoor work areas or on small vessels may not be practical but employers must take these provisions into consideration.

Best practice guidance

- Provide toilet facilities, where possible, and ensure that they are kept clean, are private and have light, heat and ventilation.
- Where toilet facilities are not practical, consider placing a portable toilet or chemical toilet in a convenient location, such as the dock.
- Never permit the unsanitary act of urinating or defecating into the water.
- Provide adequate breaks to allow for the use of toilet and washing facilities.
- Provide adequate hand-cleaning facilities, such as soap and water or hand sanitizer.
- Provide emergency eyewash facilities where poisonous, irritating or infectious material may be used. When using hazardous products, follow the recommendations on the Safety Data Sheet (SDS) for first aid.
- Shower facilities, where available, must be adequately supplied and maintained.

5.4 Personal Protective Equipment (PPE)

Personal protective equipment (PPE) is an important part of the safety program. It is essentially any clothing or equipment that helps to protect a worker from an occupational injury or disease as a result of exposure to hazards such as chemicals, bioaerosols, airborne contaminants, noise and solar radiation. Workers are required to wear protective equipment *appropriate to the hazards* they are exposed to based on the adequacy of other control measures. Different forms of PPE provide appropriate protection to a specific part of the body. PPE need to fit and be worn properly for them to be effective. Employers should ensure that workers are trained regularly on correct PPE use.

Best practice guidance

- A complete OSH risk assessment and review of the operation and existing control measures will show where PPE is required.
- The employer is responsible for providing PPE if other control measures are absent or inadequate.
- Workers are required to wear the correct PPE when instructed to do so, as a condition of employment.
- Workers should be involved in the selection of appropriate PPE.
- Respirators should be fit tested, for them to be effective.
- Workers must be trained in the proper use and maintenance of their PPE.
- All PPE must be properly maintained. Any faulty or worn-out PPE must be replaced at no cost to the worker.
- Wear the following when:
 - head protection is required – hard hat
 - eye or face protection is required – protective glasses, goggles or a face shield.
 - using loud equipment or machinery – hearing protection (for noise levels exceeding 85 decibels).
 - there is a risk of inhaling chemicals, bioaerosols, particulate dusts – respirator appropriate to the airborne hazard.
 - there is a risk of punctures, abrasions or chemical exposure to the hands and arms – work gloves.
 - needing good foot traction – safety boots or footwear.
 - working outdoors – hat that will keep sunlight off the face, sunglasses to protect eyes, and broad-spectrum sunscreen.
 - working on the water –appropriate personal flotation device for each worker.

5.5 First aid and emergencies

Workers in the inland aquaculture/fisheries sector spend a lot of their time on the water and away from immediate professional medical attention. It is therefore critically important that they know basic first aid measures and are ready for any emergency situation.

General precautions:

- Make sure that enough workers are trained in First Aid.
- Record all injuries, even minor ones. Note any First Aid treatment given.
- First Aid services must be accessible at all times.
- There should be plans in place to transport an injured worker to hospital if needed.
- The First Aid kit must be adequate for the number of workers and located in the current work area.

Best practice guidance

- Every worker must know where the First Aid kit is stored.
- Every worker must know what to do if injured or experience acute distress.
- There must be sufficient First Aid kits available.
- The First Aid kits must be clearly marked and stored in an accessible area.
- Check that the First Aid kit(s) are current – make sure that the contents have not passed their expiry date and that all items used are replaced immediately.
- Ensure that adequate means of communication are available for workers that work in isolated locations (such as on the water or ice).
- Distress, boat safety and emergency equipment must be kept in a dry, easily accessible area.

Emergency planning:

Nothing beats being prepared for an emergency situation. The employer is responsible for:

- Making sure that emergency procedures (e.g., fire, swamping, person overboard) are developed and that the appropriate training is given to all.
- Ensuring that all workers are aware of the emergency plan developed, and are capable of responding to an emergency competently.

5.6 Environmental exposure monitoring

- Exposure monitoring of the work environment is conducted to determine the level of exposure to a particular hazard identified through an OSH risk assessment and/or to evaluate the effectiveness of control measures.
- Commonly measured hazards in inland aquaculture/fisheries operations include noise levels, light (brightness) levels, bioaerosols (e.g. airborne seafood proteins), chemicals (e.g. formaldehyde in the air).
- The effectiveness and adequacy of existing ventilation can also be evaluated.
- These measurements are performed by suitably qualified and certified professionals (occupational hygienists).
- The laboratory analysing the collected samples must have quality control measures in place and be certified to do the analysis.
- Occasionally the level of certain chemicals can be measured in the blood or urine of workers frequently using them (e.g. pesticides) – the samples are collected by suitably qualified doctors or nurses.
- If exposure levels are higher than the exposure limit (international or country-specific), measures must be taken to reduce exposures using the hierarchy of control measures.
- If it is not possible to reduce exposure or it will take a long time, a worker may need to be removed from the hazardous exposure in order to prevent chronic adverse health effects.

5.7 Health monitoring (medical surveillance)

- Health monitoring, also known as medical surveillance, is undertaken to monitor whether workers develop specific long-term adverse health effects because of frequent and ongoing exposure to hazards that are poorly or inadequately controlled in the workplace.
- Health assessments are also done to assess fitness to do very specific and highly hazardous forms of work in the sector and what restrictions that may be required (e.g. diving fitness, special respirator use).
- Medical surveillance is done at baseline when a worker starts working and at regular intervals

thereafter until the person stops working in the hazardous job.

- The nature of medical surveillance performed on workers is specific to the hazard workers are exposed to, which can include specific questionnaires (for work-related symptoms of a particular body system e.g. musculoskeletal pain), medical examinations (e.g. hands for dermatitis) or tests (e.g. audiograms for hearing loss, lung function test for asthma, blood test for allergy).
- The outcome of medical surveillance can result in improved focus to control the hazard, including removal of worker/s from exposure, appropriate treatment where possible, relevant counselling, relocation to an area of low or no exposure, as well as assessment of disability and future work ability.

5.8 Information, education and training

- All workers and managers must be provided with information and undergo appropriate education and training to perform their job in a manner that protects their health and safety.
- All workers and managers must be familiar with the hazards in their jobs and the associated adverse health effects.
- All workers and managers must be familiar with the OSH protocols and best practice guidance related to specific hazards in their job.
- All workers and managers must have access to safety data sheets of all chemicals that they work with.
- All workers must be compliant and follow all OSH instructions from their supervisor, including the safe operating of machinery and the use of PPE.
- Education and training should occur on commencement of the job, when there are changes to the job or work process or if a need is identified during the OSH risk assessment.
- The employer must provide time off and cover the costs of all training related activities.

5.9 Reporting, data collection and OSH information system

- All workplaces should collect data related to occupational injuries and diseases (e.g. number of fatalities, injuries or diseases, near miss incidents) on an ongoing basis in order to monitor the effectiveness of workplace control measures.
- Workers should be encouraged to timeously report all hazardous situations or incidents related to chemicals or equipment to their supervisor – this must be recorded on an ongoing basis.
- A workplace should establish an OSH information system that include regular data collection, data analysis, interpretation of results, sharing information, and linking results to protective and preventive OSH actions.
- Employers and workers need to collectively decide what data would be the most useful to collect, how the data should be collected and stored, how data should be reported, how often and to whom group data should be reported, and finally what actions should be taken once the implications of the findings are understood.

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**APPENDIX E: Member contributions to CACFish two-year autonomous budget
for 2025 and 2026**

		Wealth component			Production component			Basic Fee
MEMBER country	Contribution Total (USD)	GDP-per capita * USD	Index	USD	Production Total ** (Tonnes)	% of total	USD	USD
Armenia	30 255	8 130	5	14 318	25 000	10.79	8 736	7 200
Azerbaijan	22 307	7 140	5	14 318	2 256	0.97	788	7 200
Kyrgyzstan	17 308	2 020	1	2 864	20 729	8.94	7 244	7 200
Tajikistan	11 697	1 180	1	2 864	4 673	2.02	1 633	7 200
Türkiye	98 435	13 240	10	28 636	179 129	77.28	62 598	7 200
TOTAL	180 000		22	63 000	231 787	100.00%	81 000	36 000

* 2023 per capita Gross Domestic Product (GDP, Current Prices, USD) of the International Monetary Fund (IMF).
<https://www.imf.org/external/datamapper/NGDPDPC@WEO/OEMDC/ADVEC/WEOWORLD/AZE/TUR/TJK/KGZ/ARM/>
 [Accessed: 21 February 2025]

** 2022 inland capture and inland aquaculture production figures of FAO (FishStatJ - Software for Fishery and Aquaculture Statistical Time Series)

BUDGET AND CRITERIA		
Year	2025–2026	
Budget	180 000	
BUDGET COMPONENTS		
Wealth	63 000	35%
Production	81 000	45%
Basic fee	36 000	20%
TOTAL	180 000	100%
WEALTH CATEGORIES		
GDP		
US \$	Index	
<999	0	
1 000-3 999	1	
4 000-5 999	3	
6 000-9 999	5	
≥10 000	10	

<https://www.fao.org/fishery/en/topic/166235?lang=en> [Accessed: 21 February 2025]

APPENDIX F: Amended version of the Rules of Procedure of the Commission

RULE I

Definitions and application

1. For the purpose of these Rules, the following definitions apply:
 - Agreement: The Agreement for the Establishment of the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission adopted by the Council of FAO on 1 October 2009, at its Hundred and Thirty Seventh Session
 - Chairperson: The Chairperson of the Commission elected in accordance with paragraph 3 of Article II of the Agreement.
 - Commission: The Central Asian and the Caucasus Regional Fisheries and Aquaculture Commission.
 - Conference: The Conference of the Organization.
 - Council: The Council of the Organization.
 - Delegation: The representative and his/her alternate designated by a Member of the Commission in accordance with paragraph 1 of Article II of the Agreement and the experts, and advisers accompanying them.
 - Director-General: The Director-General of the Organization.
 - Member: A Party to the Agreement /Member of the Commission.
 - Observer: The representative of an observer nation or organization.
 - Observer Nation, or Observer Organization: A nation that is a non-Party or not a Member of the Organization, or an international organization, governmental or nongovernmental, invited to attend a session of the Commission in accordance with paragraph 3 of Article I and Article VIII of the Agreement.
 - Organization: The Food and Agriculture Organization of the United Nations.
 - Representative: The representative of a Member of the Commission as designated in accordance with paragraph 1 of Article II of the Agreement.
 - Secretary: The Secretary of the Commission appointed in accordance with paragraph 6 of Article II of the Agreement.
 - Subsidiary Bodies: Temporary, special or standing committees and working groups established in accordance with paragraph 2 of Article VII of the Agreement.
 - Technical Advisory Committee: The Committee of the Commission established under paragraph 1 of Article VII of the Agreement.
 - Vice-Chairpersons: The Vice-Chairpersons of the Commission elected in accordance with paragraph 3 of Article II of the Agreement.
2. Unless expressly stated otherwise, these Rules shall apply, *mutatis mutandis*, to the session of the Commission and its subsidiary bodies.
3. With regard to any matter not specially covered by these Rules of Procedure or in the event of apparent conflict the respective Rules of Procedure of the Organization shall apply.

RULE II

Sessions of the Commission and its subsidiary bodies

1. Pursuant to paragraph 4 of Article II of the Agreement, the Commission, in consultation with the Director-General, shall at each session consider and agree on the time and place for the next session in accordance with the requirements of the Commission's programmes. The Chairperson, accordingly, shall issue the announcement of the session.
2. Where the Commission, at a session is unable to agree on a time and place for the next session in accordance with paragraph 1 of this Rule, it shall nevertheless, in consultation with the Director-General, take a decision as to the calendar year in which the next session is to be held. The Chairperson, in consultation with the Director-General, is then authorized to decide on the time and place of the session subject to the approval of a majority of the Members of the Commission.
3. The Chairperson, in consultation with the Director-General, may call a special session of the Commission at the request of, or with the approval of the majority of the Members of the Commission, and decide the time and place of such a session.
4. Invitations to a session of the Commission shall be issued by the Secretary, on behalf of the Chairperson, not less than sixty days in advance of the date fixed for the opening of the session. Invitations to special sessions shall be issued not less than forty days in advance of the date fixed for the opening of the session.
5. Sessions of the Commission may be held at the headquarters of the Commission, of the Organization, or at an agreed location in a country which is a Contracting Party. The Secretary shall have the mandate to deal with all logistics and related technical aspects that are instrumental to the organization of the session with the hosting country.

RULE III

Representation and credentials

1. Each Member of the Commission shall appoint one representative and one alternate, who may be accompanied by experts and advisers, to represent it in the sessions of the Commission in accordance with paragraph 1 of Article II of the Agreement.
2. At each session of the Commission, the Secretary shall receive the credentials of delegations and observers. Such credentials shall conform to the standard form set by the Secretariat. Upon examination thereof the Secretariat shall report to the Commission for necessary action, including for the Commission to decide, consistent with the Agreement, on the right of the delegation to participate in the session of the Commission and in the adoption of decisions as appropriate.
3. Each Member of the Commission shall notify the Secretary, as far as possible, in advance of the session of the Commission of the names of its representative, alternate, experts and advisers who will attend.

RULE IV

Agenda

1. The agenda of each session shall include:
 - a) election, as appropriate, of the Chairperson and two Vice-Chairpersons as provided for under Article II, paragraph 3 of the Agreement;
 - b) adoption of the agenda;
 - c) a report by the Secretary on the financial and administrative affairs of the Commission;
 - d) consideration of the proposed budget;
 - e) reports of the Technical Advisory Committee (TAC) and other subsidiary bodies (taking decisions and making recommendations as appropriate);

- f) consideration of management measures and decisions as appropriate;
 - g) programme of work;
 - h) consideration of the time and place of the next session;
 - i) proposals for amendments to the Agreement and the Rules of Procedure as appropriate;
 - j) items referred to the Commission by the Conference, the Council or the Director- General of the Organization.
2. The agenda shall also include, upon approval by the Commission:
 - a) items that were not fully discussed at the previous session;
 - b) items proposed by a Member of the Commission, the Chairperson or the Secretary.
 3. The provisional agenda, following approval by the Chairperson, shall be prepared and sent by the Secretary to Members of the Commission and observers not less than sixty days before the date of the session, together with reports and documents available in connection therewith.
 4. The agenda of a special session shall consist only of items relating to the purpose for which the session was called.

RULE V

The Secretariat

1. The Secretariat shall consist of the Secretary and such staff responsible to him/her as may be determined by the Director-General.
2. The Secretary shall be appointed for a term of three years renewable for two further terms of three years each.
3. Subject to Rule V(2) above, the Secretary shall continue to perform the functions of the office of the Secretary until a successor has taken up duties.
4. The Secretary shall be responsible for implementing the policies and activities of the Commission and for carrying out the duties and perform the functions and responsibilities of the Secretary described in Annex A of these Rules and shall report thereon to the Commission.
5. In the exercise of his/her functions, the Secretary shall have direct working relations and communication with all Members of the Commission as well as with the FAO Secretariat at all levels as appropriate.
6. Communication between the Secretary and the Members of the Commission for liaison purposes and concerning the affairs of the Commission shall be channelled, as far as possible through the National Focal points designated in accordance with Rule VI. To facilitate effective communication, the Secretary may communicate with the Members of the Commission through Permanent Representations of the Members of the Commission in Rome and FAO Representations of the Members of the Commission concerned.
7. Copies of all communications concerning the affairs of the Commission shall be sent to the Secretary for purposes of information and record.
8. The Organization may, with the approval of the Commission, provide technical support and services to the Secretariat and the work of the Commission, the Technical Advisory Committee and other subsidiary bodies. The costs for such technical support and services in accordance with this Rule shall be borne by the budget of the Commission.

RULE VI

National Focal Points

1. Each Member of the Commission shall designate a National Focal Point who shall facilitate communication between the Member of the Commission and the Secretariat. The National Focal Point shall be a high level, experienced fisheries and aquaculture scientist/manager who has access to his/her country's main fisheries and aquaculture data and information. The National Focal Point shall normally be the Representative of the Member of the Commission at the Commission sessions or any other person so designated. The name of the National Focal Point shall be communicated to the Chairperson and Secretariat of the Commission, as shall any change in that designation.
2. The National Focal Point shall perform the duties and responsibilities in the Terms of Reference described in Annex B which may be reviewed and revised from time to time by the Commission.

RULE VII

Plenary sessions and meetings of the Commission and its subsidiary bodies

1. Subject to Rule XIII and in accordance with paragraph 3 of Article I and Article VIII of the Agreement, sessions of the Commission shall be open to Observers. When the Commission decides to hold a private meeting, it shall at the same time determine the scope of such a decision with respect to observers.
2. Subject to Rule XIII, the meetings of the Technical Advisory Committee and other subsidiary bodies which may be established shall be open only to Delegations unless otherwise decided by the Commission.

RULE VIII

Election of Chairperson and Vice-Chairpersons

1. Subject to paragraph 4, the Commission shall, during each session, or sooner where an office becomes vacant, elect the Chairperson and two Vice-Chairpersons of the Commission pursuant to paragraph 3 of Article II of the Agreement, who shall assume office immediately following the session at which they were elected.
2. Nominees for the Chairperson or Vice-Chairpersons must be delegates or alternates of Members attending the session.
3. The Chairperson and Vice-Chairpersons shall serve for a term of two years. The Chairperson and Vice Chairpersons shall be eligible for re-election, but they may not be elected to serve for more than two consecutive terms.
4. Notwithstanding paragraph 1, the Chairperson and Vice-Chairpersons of the Commission appointed at its inaugural session shall assume their roles immediately upon their election.

RULE IX

Functions of the Chairperson and Vice-Chairpersons

1. The Chairperson shall exercise the functions conferred on him/her by the Agreement and in these Rules and, in particular, shall:
 - a) declare the opening and closing of each plenary meeting of the Commission;
 - b) direct the discussions at such meetings and ensure observance of these Rules, accord the right to speak, put questions and announce decisions;
 - c) rule on points of order;
 - d) subject to these Rules, have complete control over the proceedings of the session;
 - e) appoint such ad hoc, temporary, special and standing committees of the session as the Commission may direct;

- f) call for votes and announce results of the votes;
 - g) sign on behalf of the Commission a report of the proceedings of each session of the Commission for transmission to the Director General and the Members of the Commission;
 - h) perform any function as may be decided by the Commission.
2. In the absence of the Chairperson, or at his request, his functions shall be exercised by the first Vice-Chairperson or, in the absence of the latter, by the second Vice-Chairperson.
 3. The Chairperson or Vice-Chairpersons, when acting as Chairperson shall not vote and another member of their delegations shall represent their governments.
 4. The Chairperson or the Vice-Chairperson acting as Chairperson has a right to vote if he is acting as the only representative of his country.
 5. In the interval between sessions of the Commission, the Chairperson shall exercise the functions assigned to him/her by the Agreement.
 6. The Secretary shall temporarily exercise the functions of the Chairperson in the event the Chairperson and Vice-Chairpersons are unable to serve.

RULE X

Quorum, Decisions, Voting and Procedures

1. A majority of the total membership of the Commission shall constitute a quorum at any session of the Commission.
2. In accordance with paragraph 2 of Article II of the Agreement, each Member of the Commission shall have one vote.
3. Except as otherwise expressly provided for in the Agreement, all decisions of the Commission shall be taken by a majority of the votes cast.
4. The Commission shall endeavour to reach its decisions by consensus without having to resort to formal voting.
5. Decisions of the Commission on the matters described in paragraphs 7 and 8 of Article II, paragraph 1 of Article V, paragraph 2 of Article IX and Article XII of the Agreement shall be taken in accordance with the terms of those Articles.
6. Except as provided in paragraph 4 of this Rule, voting in plenary meetings shall be oral or by show of hands, except that a vote by roll call shall be taken if a specified majority is required by the Agreement or these Rules, or if a request for a vote by roll call is made by any delegation.
7. A vote by roll call shall be conducted by calling upon delegations in the English alphabetical order.
8. The record of any roll call vote shall show the votes cast by each delegate and any abstention.
9. Voting and the election of officers of the Commission, its Technical Advisory Committee and other subsidiary bodies, shall be by secret ballot except that in the case of an election in which there are not more candidates than vacancies the Chairperson may submit to the Commission that the appointment be decided by clear general consent.
10. When no nominee for an office obtains on the first ballot a majority of the votes cast, there shall be taken a second ballot confined to the two candidates obtaining the largest number of votes. If, on the second ballot, the votes are equally divided, the Chairperson shall decide between the candidates by drawing lots.
11. If the Commission is equally divided when a vote is taken on a question other than an election, a second vote shall be taken at the next meeting of the current session. If the Commission is then again equally divided, the proposal shall be regarded as rejected.

12. Voting arrangements and other related matters not specifically provided for by the Agreement, or by these Rules, shall be governed mutatis mutandis by the provisions of the General Rules of the Organization.

13. The right to vote of a Member of the Commission who is in arrears shall be subject to the paragraph 7 of Article IX of the Agreement.

RULE XI

The Technical Advisory Committee and other subsidiary bodies

1. The Technical Advisory Committee shall have the functions and responsibilities described in the Terms of Reference in Annex C of these Rules.

2. The establishment of subsidiary bodies shall be subject to the provisions of Article VII, paragraph 3 of the Agreement.

3. The procedures of the Technical Advisory Committee and other subsidiary bodies shall be governed mutatis mutandis by the Rules of Procedure of the Commission.

RULE XII

Budget and finance

1. The Commission shall administer its budget and other funds pursuant to its Financial Regulations adopted in accordance with paragraph 8 of Article II and Article IX of the Agreement.

2. The Financial Regulations of the Organization, as implemented by the Administrative Manual and memoranda and the procedures based thereon, shall, apply to the Commission until such time the Commission adopts its financial regulations in accordance with paragraph 8 of Article II of the Agreement.

RULE XIII

Participation by Observers and Others

1. The Director-General or a representative designated by him/her, shall have the right to participate without vote in all meetings of the Commission, of the Technical Advisory Committee and of other subsidiary bodies of the Commission.

2. Members of the Organization that are non-Members of the Commission may, upon their request, be invited to be represented by an observer at sessions of the Commission in accordance with the terms and conditions established by the Commission.

3. Non-Members of the Commission and non-Members of the Organization that are Members of the United Nations, any of its Specialized Agencies or the International Atomic Energy Agency, may, upon request made to and granted by the Commission in accordance with paragraph 3 of Article I of the Agreement and the principles relating to the granting of observer status to Nations adopted by the Conference as well as the terms and conditions established by the Commission, be invited to attend sessions of the Commission and its subsidiary bodies in an observer capacity.

4. The Commission may, on their request or on the proposal of the Secretary pursuant to Article VIII of the Agreement, invite international organization, governmental or nongovernmental having special competence in the field of activity of the Commission, to attend such of its meetings as the Commission may specify and in accordance with the terms and conditions established by the Commission. The list of the nongovernmental organization wishing to be invited in accordance with this rule shall be submitted beforehand by the Secretary to the Members of the Commission. If one of the Members of the Commission objects giving in writing its reasons within 30 days, the matter will then be subject to decision of the Commission out of session by written procedure.

5. Participation of regional economic integration organizations in the work of the Commission and the relations between the Commission and such organizations shall be governed by the relevant provisions of the Constitution and the General Rules of the Organization, as well as by the rules on the relations with international regional economic integration organizations adopted by the Conference or the Council.

6. Unless the Commission expressly determines otherwise, observers may attend the plenary meetings of the Commission. Observer members of the Organization may submit memoranda and participate in the discussions. Observer Non-Members of the Organization as well as Observer intergovernmental organizations may be invited by the Commission to submit memoranda and deliver oral statements.

7. The Commission may invite consultants or experts, in their individual capacity, to attend the meetings or participate in the work of the Commission as well as the Technical Advisory Committee and the other subsidiary bodies of the Commission in accordance with the terms and conditions established by the Commission.

8. In no case will observers and persons invited to attend the meetings or participate in the discussions of the Commission be entitled to vote.

9. In accordance with Article VIII of the Agreement, the Commission may enter into agreements with other intergovernmental organizations and institutions, especially those active in the fisheries and aquaculture sector.

RULE XIV

Records, reports and recommendations

1. A written report shall be adopted at the end of each session of the Commission. The report shall be published together with such technical papers and other documents as the Commission may wish to publish.

2. The report adopted pursuant to paragraph 1 of this Rule shall, in accordance with Article VI of the Agreement, embody the Commission's views, recommendations and decisions, including, when requested, a statement of minority views. Such report, together with the schedule for implementation of decisions and recommendations by the Members of the Commission as appropriate, shall be transmitted to the Director-General.

3. The Secretary shall circulate the report as well as such technical papers and other documents as appropriate, to the Members of the Commission and other States and international organizations that were represented at the session or that have cooperative arrangements with the Commission. These documents may be made available to other Members and Associate Members of the Organization for their information, as appropriate.

4. Decisions and recommendations which might have policy, programme or financial implications for the Organization shall be brought by the Director-General to the attention of the Conference through the Council for action.

5. The Chairperson may request each Member of the Commission to provide information to the Commission or the Director-General on actions it has taken and where such actions are based on or related to the decisions and recommendations of the Commission.

RULE XV

Amendments to the Agreement

1. Proposals for the amendment of the Agreement, as provided by Article XII of the Agreement, shall be communicated in writing to the Secretary. The Secretary shall transmit to all Members of the Commission and to the Director-General a copy of such proposals immediately upon their receipt.

2. No action on a proposal for the amendment of the Agreement shall be taken by the Commission at any session, unless it has been included in the provisional agenda of the session. Decisions on amendments shall be made in accordance with Article XII of the Agreement.

RULE XVI

Suspension and amendment of Rules

1. Subject to the provisions of the Agreement, any of the foregoing Rules, other than Rules IV, V, XI paragraph 2, XII, XIII, may be suspended by a two-thirds majority of the votes cast at any plenary meeting of the Commission, provided that announcement is made at a plenary meeting of the Commission and copies of the proposal for suspension have been distributed to the delegations not less than forty-eight hours before the meeting at which action is to be taken.

2. Amendments of, or addition to, these Rules may be adopted by a two-thirds majority of the Members of the Commission, at any plenary meeting of the Commission, provided an announcement is made at a plenary meeting and copies of the proposal for the amendment or addition have been distributed to delegations not less than twenty-four hours before the meeting at which action is to be taken.

3. Any amendment to Rule XVII which may be adopted in accordance with the provisions of paragraph 2 of this Rule, shall not become effective until the next session of the Commission.

RULE XVII

Languages of the Commission

1. The official language of the Commission shall be English.

2. During the meetings, interpretation in Russian shall be provided, as appropriate, by the host country. A delegation using a language other than English or Russian shall provide, at its own expenses, for interpretation, translation and publication, as appropriate, into the official language.

3. The reports of the Commission and its subsidiary bodies will be made available in English and Russian by the Secretariat. The Commission, in deciding to accept submissions of publications or reports and communications in a language other than English or Russian, may request an abstract of the publication, report and communication, as appropriate, to be translated into the official language.

Functions and responsibilities of the Secretary

In addition to the functions and responsibilities assigned to the Secretary by the Agreement and these Rules of Procedures, the duties of the Secretary shall include:

- a) communication of information received from Members of the Commission;
- b) receipt, collection, circulation, drafting and presentation of documents, reports, papers and resolutions for the sessions of the Commission, the Technical Advisory Committee and other subsidiary Committees and Working Groups;
- c) maintaining records of the proceedings of sessions of the Commission, the Technical Advisory Committee and other subsidiary bodies;
- d) facilitating the collection of data necessary to accomplish the objectives of the Commission;
- e) administering and reporting to the Commission on the financial and staffing resources of the Commission, including the certification of expenditures and financial commitments;
- f) maintain contacts with appropriate government officials, fisheries and aquaculture institutions and international organizations concerned with the development, conservation, rational management and utilization of fisheries, as well as the sustainable development of aquaculture in the area of application, to facilitate consultation and cooperation on all matters pertaining to the objectives of the Commission;
- g) subject to Rule II of these Rules of Procedure deal with the practical arrangements relating the sessions of the Commission and meetings of its subsidiary bodies and other related ad hoc meetings;
- h) prepare, or arrange for the preparation of, background documents and papers and a report on the Commission's activities and the programme of work for the submission to the Commission at its sessions and arrange for the subsequent publication of the report and the proceedings of the Commission as well as its subsidiary bodies and related ad hoc meetings;
- i) performance of such other duties as may be decided by the Commission.

Terms of reference for National Focal Points

The National Focal Point designated in accordance with Rule VI shall facilitate liaison and communication between the Commission and its Members. He/She shall:

- a) keep in direct contact with the Secretariat of the Commission;
- b) receive information from the Secretariat of the Commission on activities planned and implemented at the regional level;
- c) Inform the Secretariat on all the activities carried out at national level relating to the work of the Commission;
- d) advise the Commission about appropriate scientific and other research programmes, including possible annual revisions, within the limits of the available project budget;
- e) review and discuss the annual work plans offering proposals for and planning of scientific, training and other courses and seminars in the context of the Commission work programme and activities;
- f) support liaison and coordination on scientific, technical and other matters with relevant fisheries and aquaculture organizations or related projects and with appropriate institutions and interested scientists within the jurisdiction of the Members of the Commission or in the region;
- g) coordinate, at the national level, specific activities and initiatives of the Commission, and provide all the necessary support in the acquisition of information and data;
- h) ensure effective flow of communications regarding the Commission's activities to and from national scientific, institutional and other relevant counterparts and stakeholders.

Terms of reference of the Technical Advisory Committee (TAC)

The Technical Advisory Committee shall be composed of scientists or technical experts on inland fisheries and aquaculture research and development. The member of the Technical Advisory Committee may be from an appropriate government agency or from a university or a scientific institute.

The Technical Advisory Committee shall be open to all Members of the Commission. Each Member of the Commission shall designate a representative to represent the Member of the Commission in the Technical Advisory Committee. Such representative may be accompanied by other experts and advisers.

The Technical Advisory Committee shall have the following duties and responsibilities:

- a) work towards operationalization of policies and decisions of the Commission;
- b) formulate and provide technical and scientific advice to the Commission and its work;
- c) propose the work programme of the Commission with the assistance of the Secretariat;
- d) provide independent technical and scientific advice and expert inputs to assist the Commission in implementing its work programme and in monitoring and evaluating such implementation;
- e) provide technical oversight, monitoring and evaluation of projects and programme of work activities;
- f) provide advice on the technical and scientific bases for decisions and recommendations on measures concerning general conservation and rational management of living aquatic resources, including the biological, social and economic aspects of fisheries and aquaculture management;
- g) propose the establishment of subsidiary bodies to work on specific technical issues or to provide a specified output;
- h) provide information on fisheries and aquaculture production and other data relevant to the functions of the Commission in such a way as to enable the Commission to achieve its objectives.

The Ninth Session of the Central Asian and Caucasus Regional Fisheries and Aquaculture Commission (CACFish) was held in Almaty, Kazakhstan, on 25–26 November and hosted by the Fisheries Committee of the Ministry of Agriculture of Kazakhstan. The meeting brought together four Member States: Azerbaijan, Kyrgyzstan, Tajikistan and Türkiye, along with five participating non-Member States: Georgia, Kazakhstan, the Russian Federation, Turkmenistan and Ukraine. The Commission reviewed key decisions and recommendations from FAO Regional Conferences, Committees and Regional Fisheries Bodies, emphasizing the importance of ensuring that regional priorities are reflected in global and regional policy processes. The Secretariat presented administrative and financial updates, noting improved compliance by one Member State while persistent arrears from two others continue to challenge financial sustainability. Membership expansion was identified as a positive development, with additional countries expected to join as accession processes advance. Technical discussions focused on adapting FAO Guidelines for Sustainable Aquaculture to the regional context, advancing zoning and site selection, developing a fisheries data collection system and adopting non-binding Best Practice Guidance on Occupational Safety and Health. The Commission acknowledged its expanding technical scope and achievements during the intersessional period, reaffirming its role in addressing regional challenges. The Session approved an autonomous annual budget of USD 180 000 for 2025–2026, endorsed revisions to the Rules of Procedure and adopted the work programme for 2025–2027.

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